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Effects of Baduanjin Exercise on Motor Function, Balance and Gait in Parkinson's Disease: A Systematic Review and Meta-analysis

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Effects of Baduanjin Exercise on Motor Function, Balance and Gait in Parkinson's Disease: A Systematic Review and Meta-analysis

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ABSTRACT

Objective This study aims to systematically evaluate the effects of Baduanjin on motor function, balance and gait in Parkinson's disease (PD) patients.

Design Systematic review and meta-analysis.

Study selection We included all eligible randomized controlled trials (RCTs) (published in English and Chinese).

Data sources The following 10 electronic databases were systematically searched, from inception to March 17, 2022: PubMed, Web of Science, Cochrane Library, Embase, EBSCOhost, OVID, SinoMed, China National Knowledge Infrastructure (CNKI), Wanfang Data, and China Science Journal Database (VIP).

Review methods Methodological quality assessment and meta-analysis were performed for the included studies using the Cochrane Review Manager Version 5.4 software.

Results A total of 10 randomized controlled trials (RCTs) with 808 participants were included. The results revealed the following: (1) Baduanjin significantly improved the motor function of PD patients, based on the Unified Parkinson's Disease Rating Scale, Part III (UPDRS-III; MD = -5.37, 95% CI: -8.96 to -1.78, P=0.003) and Fugl-Meyer Assessment of Lower Extremity (FMA-LE) (MD = 5.39, 95% CI: 2.71 to 8.07, P<0.0001); (2) Baduanjin significantly improved the ability of balance of PD patients, based on the Berg Balance Scale (BBS) (MD = 4.40, 95% CI: 3.08 to 5.73, P<0.00001); (3) Baduanjin significantly improved the gait of PD patients, based on the 6-minute walk distance (6MWD) (MD = 21.62, 95% CI: 11.14 to 32.10, P<0.0001). After further subgroup and sensitivity analyses, the heterogeneity was

identified to be potentially due to the different degrees of disease severity in PD patients and the differences in Baduanjin intervention duration.

Conclusions The analysis of this systematic evaluation indicates that Baduanjin can significantly improve the motor function, balance and gait of PD patients, when compared to PD patients without Baduanjin intervention. However, this conclusion still warrants more high-quality and multi-center RCTs for further verification.

Keywords Baduanjin; Parkinson's Disease; Motor Function; Balance; Gait; Meta-analysis

Strengths and limitations of this study

- First systematic review to assess effectiveness of effects of Baduanjin on motor function, balance and gait in Parkinson's disease (PD) patients.
- Two reviewers independently considered the abstracts and full-text articles for inclusion, extracted the data, and assessed the included studies for risk of bias in the same way.
- Our confidence in the findings is increased by sensitivity analyses and further subgroup analyses.
- The findings are likely to be influenced due to the bias in some original trials.

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INTRODUCTION

Parkinson's disease (PD) represents the second most common neurodegenerative disease, which is characterized by the decrease in dopamine concentration in the basal ganglia and the decrease in dopaminergic neurons in the substantia nigra pars compacta (SNpc) [1]. According to the global survey of nervous system diseases, the rate of incidence and prevalence of PD have been dramatically increasing. In addition, this has been recognized as the potentially fastest-growing neurological disease in the world [2,3]. From 1990 to 2015, more than six million people suffered from PD [4]. Considering the increasing aging population, the number of PD patients is expected to reach more than 12 million by 2040, which would be almost double the present number [5]. PD cardinal clinical symptoms mainly include resting tremors, postural balance disorder, bradykinesia and rigidity [6]. Furthermore, PD patients are more prone to falls or fractures due to balance disorder and abnormal gait, limiting their activity ability and seriously affecting their quality of life (QoL), and imposing a heavy burden on families and society. Hence, the timely prevention or treatment of PD with effective strategies is imperative.

At present, levodopa remains as the main and first-line drug for the clinical treatment for PD patients [7]. Although drug therapy can improve the patient's symptoms, the effectiveness of pharmacological treatment tends to wear off over time, resulting in severe motor fluctuations and dyskinesia [8]. Furthermore, motor complications of any kind can severely impact motor control, and lead to increased disability and decreased QoL [9]. Given the high incidence of PD, the progressive nature of the disease, and the short duration of drug effectiveness, a pursuit for complementary and alternative medicine (CAM) would be essential to improve the QoL of patients, and minimize the burden on families and society [10]. The worldwide interest in using CAM to treat PD patients is growing. CAM include supplements, massage therapy, physical activity and mind-body exercises [11].

Baduanjin is a traditional mind-body exercise that originated from the Song Dynasty, and has existed for more than one thousand years. This consists of eight simple movements, in which the whole set of activities are gentle, slow and continuous. This is characterized by deep and slow breathing, emphasizing static and dynamic posture control, and combining body movement and breathing patterns [12]. Baduanjin exercises can improve the patient's ability to focus on postural stability, strengthen core muscles, put pressure on weight-bearing joints, increase

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proprioception, and control body posture [13]. In addition, Baduanjin is often referred to as a moderate-intensity aerobic exercise that is easy to learn and practice at home. Based on its characteristics, this is very suitable for special populations, including physical disabilities, such as PD patients with relatively low tolerance to physical activity intensity, and good compliance with chronic disease prevention and treatment [14]. In recent years, a number of trials have been conducted to investigate the effects of Baduanjin on PD. However, to date, no systematic evaluation of these research results has been identified. Therefore, in order to determine the effectiveness of Baduanjin in PD, a systematic review and meta-analysis of Baduanjin randomized controlled trials (RCTs) on PD patients was conducted, particularly in terms of the motor function, balance and gait of PD patients.

METHODS

The present study was conducted in accordance with the guidelines of the Cochrane Handbook (PRISMA) [15].

Patient and public involvement

No patient involved.

Search Strategy

An electronic literature search was performed on the following databases for Chinese and English language articles published as of March, 2022: PubMed, Web of Science, Cochrane Library, Embase, EBSCOhost, OVID, SinoMed, China National Knowledge Infrastructure (CNKI), Wanfang Data, and China Science Journal Database (VIP). The following search terms were used: Baduanjin, Mind-body exercise, Qigong, and Parkinson's disease. Taking the PubMed database as an example, the search strategy is provided in the appendix.

Inclusion and Exclusion Criteria

The inclusion criteria for the present study were, as follows: (1) studies that targeted PD patients; (2) RCTs; (3) the control group received other treatments (*e.g.* drug therapy, balance board training, conventional treatment, and conventional physical therapy), while the experimental group received Baduanjin exercises, in addition to the interventions used in the control group; (4) for studies with multiple

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groups, all groups that met the above criteria were included; (5) outcome indicators: types correlated to motor function, balance and gait.

The exclusion criteria for the present study were, as follows: (1) the experimental group received Baduanjin exercises combined with another intervention not used in the control group (*e.g.* Baduanjin combined with balance board training); (2) duplicate publications; (3) reviews, animal experiments, experience reports, etc.; (4) studies with no relevant results.

Study Selection and Data Extraction

Two researchers independently screened the included studies and extracted the relevant data. The extracted data included the general information (first author and publication year, location, Hoehn and Yahr grade, age and sample size), details of the interventions (interventions measured in the control and Baduanjin groups, frequency of interventions, and intervention duration), and outcome indicators. Any disagreement during the process was resolved by discussion with a third researcher for confirmation.

Risk of Bias

Two researchers used the bias risk assessment tool, which is applicable for RCTs and recommended by the Cochrane Manual 5.1.0, in order to independently evaluate the included literature and cross-check the results. If there was a disagreement, a third researcher would arbitrate to reach a consensus. The risk of bias evaluation involved seven aspects: random sequence generation, allocation concealment, blinding of researchers and participants, blinding of evaluators, completeness of outcome indicators, selective reporting, and other bias.

Statistical Analysis

The Cochrane Review Manager version 5.4 software was used for the data analysis. The count data were presented in odds ratio (OR) and 95% confidence intervals (CIs), and the measurement data was presented in mean difference (MD). The I^2-test was applied to estimate the statistical heterogeneity among trials. Studies with $I^2 \leq 50\%$ were considered to have lower heterogeneity, and these were analyzed using a fixed effect model. Otherwise, significant heterogeneity was considered among the studies, and the random effects model was adopted. Then, a subgroup

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analysis (e.g. according to the duration of interventions) or sensitivity analysis was performed, and the possible reasons for the high heterogeneity were explored. The difference was defined to be statistically significant when $P<0.05$.

RESULTS

Search Results

A total of 267 articles were retrieved from the databases. Among these articles, 125 duplicate articles were excluded. Then, the titles and abstracts of the remaining 142 articles were screened, and 125 articles that were not qualified based on the criteria were excluded. After the full-text screening of the remaining 17 articles, a total of seven articles that did not meet the study criteria were excluded. The reasons for the exclusion were, as follows: non RCTs (n=4), clinical case reports (n=2), and duplicate publications (n=1). Finally, a total of 10 studies were included [16-25] for the present meta-analysis. The flowchart for the study selection is presented in Figure 1.

Characteristics of the Included Studies

The present review included at total of 10 [16-25] RCTs with 808 participants diagnosed with PD. For the language used by the included studies, eight [16-23] studies were reported in the Chinese language, and two [24-25] studies were reported in the English language. Furthermore, for the 10 included RCTs, five RCTs [16,18,21,22,25] clearly reported the disease grading of PD patients, while five RCTs [17,19,20,23,24] did not clearly report this. The control group received drug therapy, balance board training, conventional physical therapy, etc. The experimental group received any of the interventions in the control group plus Baduanjin. The Baduanjin intervention in the included studies had a frequency that ranged within 4-7 times a week and a duration that lasted for 1-24 weeks. The main outcome indicators included the following: (1) tests of motor function, including the Unified Parkinson's Disease Rating Scale, Part III (UPDRS-III), Fugl-Meyer Assessment of Lower Extremity (FMA-LE), and Function Independent Measure (FIM); (2) indicators of balance, including the Berg Balance Scale (BBS) and Balance-check® balancer test; (3) indicators of gait, including the 6-minute walk distance (6MWD), Timed Up and Go Test (TUGT), Intelligent Device for Energy Expenditure and Activity gait (IDEEA gait), Tinetti Performance Oriented Mobility Assessment (Tinetti POMA), 10-meter Walk Test

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(10MWT), and Freezing of Gait (FOG). The characteristics of the analyzed trials are presented in Table 1.

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Table 1. Basic characteristics of the included studies

Study	Location (language)	Hoehn & Yahr stage	Age	Sample		Intervention		Frequency	Duration (Week)	Outcomes
				C	E	C	E			
Dong SS, 2022[16]	Jiangsu, China (Chinese)	1-3	C: 63.07±12.78 E: 65.37 ± 7.47	30	27	Drug therapy	Drug therapy+Baduanjin	30min/session, 1 session/d, 7d/wk	3	IDEEA gait, Balance-check® balancer test
Shi XW, 2021[17]	Henan, China (Chinese)	Not reported	60-75	64	65	Drug therapy+Balance board training	Drug therapy+ Balance board training+Baduanjin	60min/session, 1 session/d, 4d/wk	8	FMA-LE, 6MWD, UPDRS-III, BBS
Song ZX, 2020[18]	Jiangsu, China (Chinese)	1-3	>65	60	60	Drug therapy+Balance board training	Drug therapy+ Balance board training+Baduanjin	20min/session, 1 session/d, 5 d/wk	8	FMA-LE, 6MWD, UPDRS-III, BBS
Zhang XY, 2020[19]	Fuzhou, China (Chinese)	Not reported	61-74	41	42	Drug therapy+Balance board training	Drug therapy+ Balance board training+Baduanjin	30min/session, 1 session/d, 7 d/wk	12	6MWD, FIM
Zhou SY, 2020[20]	Changsha, China (Chinese)	Not reported	60-86	39	39	Balance board training	Balance board training+Baduanjin	20min/session, 1-2session/d, 5 d/wk	4	FMA-LE, 6MWD
Zhi X, 2020[21]	Dalian, China (Chinese)	1-2	45-65	15	15	Drug therapy	Drug therapy+ Baduanjin	60min/session, 1 session/d, 6d/wk	12	UPDRS-III, TUGT, 10MWT
Wei N, 2019[22]	Beijing, China (Chinese)	1-3	32-78	22	28	Drug therapy+Health education	Drug therapy+Health education+Baduanjin	60min/session, 1 session/d, 7d/wk	During hospitalization (1-2w)	UPDRS-III
Yang D, 2019[23]	Chongqing, China (Chinese)	Not reported	45-82	50	50	Conventional treatment	Conventional treatment+Baduanjin	Not reported	8	BBS, Tinetti POMA
Xiao C, 2016[24]	Beijing, China (English)	Not reported	67.8 ± 9.4	33	32	Conventional physical therapy	Conventional physical therapy+Baduanjin	60min/session, 1 session/d, 7d/wk	24	BBS, TUGT, 6MWD, FOG, UPDRS-III
Xiao CM, 2015[25]	Beijing, China (English)	1-3	55-80	48	48	Walking	Walking+Baduanjin	60min/session, 1 session/d, 4d/wk	24	UPDRS-III, BBS, TUGT, 6WMD

Note: C=Control group; E=Experiment group (Baduanjin group) **Outcomes:** IDEEA gait, Intelligent Device for Energy Expenditure and Activity gait; FMA-LE, Fugl-Meyer Assessment of Lower Extremity; 6MWD, 6-minute Walking Distance; FIM, Functional Independence Measurement; UPDRS-Ⅲ, Unified Parkinson's Disease Rating Scale, Part Ⅲ; BBS, Berg Balance Scale; TUGT, Timed Up and Go Test; 10MWT, 10-meter Walk Test; POMA, Performance Oriented Mobility Assessment; Tinetti POMA, Tinetti Performance Oriented Mobility Assessment; FOG, Freezing of Gait

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Methodological Quality and Risk of Bias

The methodological quality and risk of bias for each included study are presented in Figures 2 and 3. Four studies [16,17,19,20] reported the random sequence-generation method, while one study [25] used the blinded outcome assessors. All these studies were judged to have a low risk of bias in the aforementioned domains. In addition, one study [22] was grouped according to the way patients voluntarily participated. Hence, this was rated as "high risk". All selective reporting and other bias risks included in the trial were considered as "low risk".

Meta-analysis

Effect of Baduanjin on Motor Function

UPDRS-Ⅲ

UPDRS-III is part of the standard clinical test to assess the severity of motor complications in PD [26]. Five studies [17,18,21,22,25] with 418 participants used UPDRS-III to evaluate the motor symptoms. The meta-analysis revealed that the UPDRS-III scores were lower in the Baduanjin group, when compared to the control group (MD = -5.37, 95% CI: -8.96 to -1.78, P=0.003, I^2=94%; Fig. 4a). The heterogeneity sensitivity analysis was performed after excluding the study conducted by Shi *et al.* [17] The subsequent meta-analysis yielded the following results: MD = -3.40, 95% CI = -4.15 to -2.64, P<0.00001, I^2=24% (Fig. 4b). The heterogeneity may be due to the different degrees of PD among participants, and merely the study conducted by Shi *et al.* did not report the stages of the participant's PD. These above results indicate that Baduanjin can improve the degree of motor function impairment in PD.

Furthermore, subgroup analyses were performed, according to the intervention durations. The meta-analysis of two studies [18,22] with a Baduanjin intervention duration of ≤8 months revealed the following: MD = -2.83, 95% CI = -3.82 to -1.83, P<0.00001, I^2=0%. The meta-analysis of two studies [21,25] with a Baduanjin intervention duration of >8 months revealed the following: MD = -4.17, 95% CI = -5.32 to -3.01, P<0.00001, I^2=0% (Fig. 4c). These results indicate that Baduanjin can reduce the degree of motor function impairment in PD, regardless of whether the intervention duration is ≤8 weeks or >8 weeks.

FMA-LE

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FMA-LE is a frequently used injury-based measure to assess the motor recovery in the lower extremity and predict the functional recovery of individuals with PD [27]. Three studies [17,18,20] that involved 313 participants used FMA-LE to assess the lower extremity motor function. The meta-analysis revealed that the FMA-LE scores were higher in the Baduanjin group, when compared to the control group (MD = 5.39, 95% CI = 2.71 to 8.07, P<0.0001, I^2=92%; Fig. 5a). Furthermore, a heterogeneity sensitivity analysis was performed after excluding the study conducted by Song *et al.* [18], and the subsequent meta-analysis results were consistent, with the following results: MD = 6.64, 95% CI = 5.69 to 7.59, P<0.00001, I^2=39% (Fig. 5b). The heterogeneity among these studies may be attributable to the difference in Baduanjin intervention duration. These results indicate that Baduanjin can improve the motor function of the lower limbs in PD patients.

Effect of Baduanjin on Balance Function

BBS

BBS is a test used to assess functional balance, showing high validity and reliability in PD [28]. Three studies [17,18,23] that included 318 participants used BBS to evaluate the balance function. The meta-analysis revealed that the BBS scores were higher in the Baduanjin group, when compared to the control group (MD = 4.40, 95% CI = 3.08 to 5.73, P<0.00001, I^2=87%; Fig. 6a). A heterogeneity sensitivity analysis was performed after excluding the study conducted by Shi *et al.* [16], and the consistent results were, as follows: MD = 3.71, 95% CI = 2.90 to 4.51, P<0.00001, I^2=63% (Fig. 6b). These results indicate that Baduanjin has a positive effect on balance function in PD.

Effect of Baduanjin on Gait

6MWD

The 6MWD is a simple method to assess gait by instructing patients to walk for six minutes, and recording the longest distance the subject could walk [29]. Five studies [17-20,25] that involved 485 participants used the 6MWD to evaluate the walking ability. The results revealed that the 6MWD scores were higher in the Baduanjin group, when compared to the control group (MD = 21.62, 95% CI = 11.14 to 32.10, P<0.0001, I^2=80%; Fig. 7a). After removing the studies conducted by Zhou *et al.* [20] and Xiao *et al.* [25], the heterogeneity sensitivity analysis and subsequent meta-analysis revealed

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similar results: MD = 18.86, 95% CI = 13.69 to 24.02, $P<0.00001$, $I^2=0\%$ (Fig. 7b). The heterogeneity may have been attributed to the differences in the duration of Baduanjin interventions between the two studies [20,25] and other studies. These results indicate that Baduanjin can improve gait in PD.

Descriptive Analysis

One study [24] presented the data in median (range). Hence, no combined analysis was conducted. The results of this study revealed that Baduanjin can improve the balance and lower limb strength of PD patients. Furthermore, two studies [21,24] used TUGT to evaluate the gait. However, due to the high heterogeneity and small number of pooled studies, the source of heterogeneity could not be identified. Hence, the meta-analysis was not conducted. One study [16] used IDEEA gait, and other study [23] used Tinetti POMA. In addition, one study [24] used FOG, and another one study [21] used the 10MWT. All these methods were used to assess the gait function, and the results suggest that Baduanjin can improve the gait function of PD patients. In addition, the study conducted by Zhang *et al.* [19] used FIM to assess the functional independence, and the results indicated that Baduanjin has a positive effect in improving the functional exercise capacity of PD patients. The study conducted by Dong *et al.* [16] used the Balance-check® balancer test to assess the balance, and revealed that Baduanjin can significantly improve the balance of PD patients.

Among the 10 included studies, no adverse events were reported, indicating that Baduanjin can be a relatively safe exercise for PD patients.

DISCUSSON

The present study revealed that Baduanjin can reduce the UPDRS-III scores, and increase the FMA-LE, BBS and 6MWD scores, indicating that Baduanjin has a positive effect in improving the motor function, balance function, and gait of PD patients. In addition, the subgroup analysis for UPDRS-III revealed an improvement in motor function for PD patients, regardless of whether the duration of intervention was ≤8 weeks or >8 weeks.

As a chronic disease, PD has a long course, and is prone to recurrence, which is mainly caused by the progressive degeneration of dopaminergic neurons in the substantia nigra pars compacta, the decrease in striatal dopamine, and the presence of Lewy bodies [30]. Multiple abnormal neural circuits can lead to balance and gait

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problems in PD patients [31]. Gait problems are strongly dependent on impaired balance control, and the poor control of both can cause movement disorders [32,33]. Levodopa is the main drug for relieving motor symptoms of PD. However, approximately 50% of patients who receive levodopa for long durations would develop levodopa-induced dyskinesia (LID) [34]. Speck *et al.* [35] suggested through a mice model that exercise can partially prevent the development of LID, and attenuate the side effects of levodopa through the normalization of striatopallidal dopaminergic signaling, without affecting the anti-parkinsonian effect of levodopa. Hence, this indicates that Baduanjin may exhibit an effect to enhance the medication efficiency and ameliorate the side effects of levodopa, which can be used as a good CAM for the treatment of PD.

Several large controlled clinical studies have revealed that continuous exercise can improve the performance of daily activities in the early stage of PD, such as balance, gait and movement disorders [36,37], which is consistent with the present findings. The exercise guidelines for PD recommends 20-60 minutes of moderate-intensity aerobic exercise for 3-5 days a week [38]. Baduanjin, as a moderate intensity aerobic exercise, mainly comprises of eight simple left-right symmetrical movements, such as arm raising, shaking of the head, waist rotation and foot flipping [39]. Practicing Baduanjin can fully stretch the joints of the upper and lower limbs, and the movement of the trunk can improve the ability of PD patients to control their center of gravity. At the same time, Baduanjin can also improve the proprioception, strengthen the coordination and sensitivity of limbs, and promote the recovery of balance function [40]. Furthermore, Ye *et al.* [41] reported that the upper limb exercise of Baduanjin can enhance the patient's coordination, and the knee flexion and squat movement can enhance the lower limb flexor and extensor muscle strength, and lower limb weight-bearing capacity, further improving the gait ability. This explains the findings reported in the present review.

Muscle tension caused by the uncoordinated contraction of active and antagonistic muscles influence a series of motor symptoms in patients with PD [42]. There is evidence that mind-body exercises, including Tai Chi, Yoga and Qigong, can reduce the contraction of antagonist muscles [43,44]. As a kind of mind-body exercise, the long-term practice of Baduanjin can promote blood circulation in the upper and lower limbs, and lower back, strengthened the muscle metabolism and stress function, improve the synergistic contraction of active muscles and antagonist muscles, and promote the recovery of motor function [41]. The α-syn protein is the main pathogenic

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protein of PD, and its accumulation is the main reason of neuronal loss [45]. Existing studies have revealed that aerobic exercise can reduce the loss of dopaminergic neurons, increase synaptic connections, and upregulate neurotrophic factor levels to improve PD dyskinesia. Furthermore, this can downregulate the α-Syn protein levels and neuronal apoptosis, which in turn, can reduce inflammation and mitochondrial dysfunction to restore the motor function in PD patients [46-48]. Al-Jarrah *et al.* [49] proposed that aerobic exercise can decrease the level of proinflammatory proteins in the striatum in a PD experimental model by reducing the activity of microglia, thereby alleviating PD symptoms. These might be the possible mechanisms for the effect of Baduanjin in improving motor function.

Limitations and Future Prospects

The present systematic review has several limitations. First, the present study included PD patients with Hoehn and Yahr grades of below grade 3 or no reported grade. Hence, the results cannot be extended to PD patients with Hoehn and Yahr grades 4-5. Second, the present study had a relatively high heterogeneity among the literatures, such as different study intervention times, which might have been the cause for the clinical heterogeneity. Third, merely Chinese and English databases were searched. Hence, it is possible that some researches may have been missed during the database search.

CONCLUSIONS

The present review revealed that Baduanjin can effectively improve the motor function, balance function, and gait ability of PD patients. Therefore, Baduanjin exercise can be taken as a valuable aerobic exercise in clinical rehabilitation for the application of CAM.

Author contributions

Jinghui Lai, Yangfan Cai and Liyan Yang conceived and designed the study. Jinghui Lai drafted the manuscript. Yangfan Cai and Min Xia performed the data collection. Jinghui Lai, Xi Cheng and Ying Chen performed the data analysis. All authors approved the final manuscript before submitting.

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Conflicts of interest

The authors declare that they have no conflict of interest.

Data sharing statement

No additional data are available.

Ethics approval

This study does not involve human participants.

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Figure Legends

Figure 1. Literature Screening Process.

Figure 2. Risk of bias graph.

Figure 3. Risk of bias summary.

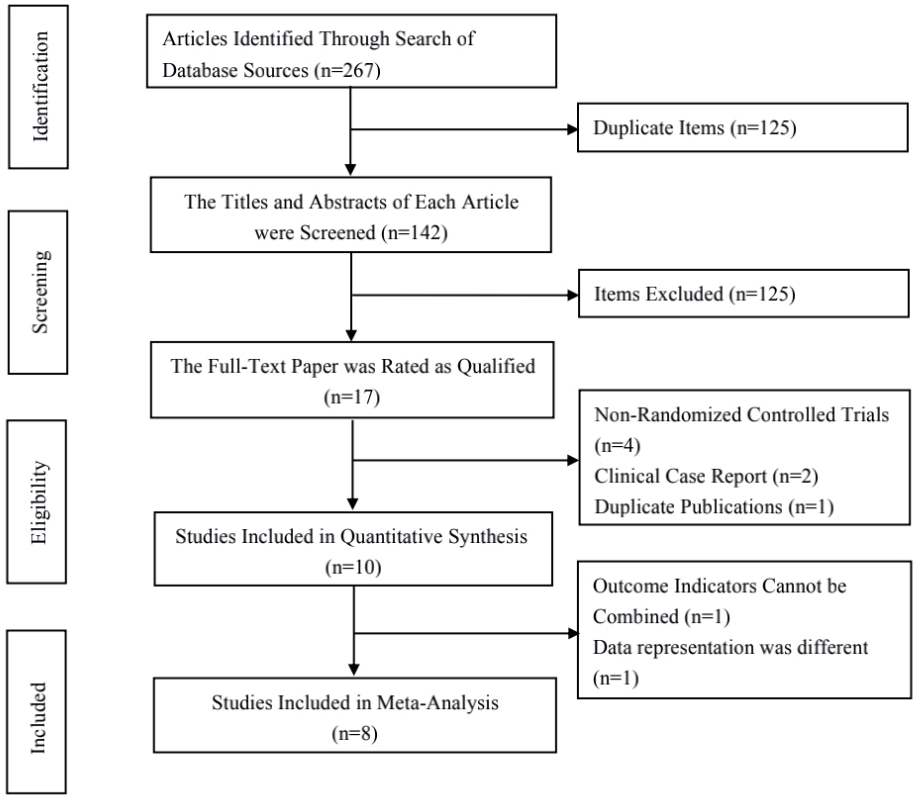
Figure 4. Total UPDRS-III scores after the Baduanjin exercise. **b** Total UPDRS-III scores after the Baduanjin exercise (sensitivity analysis). **c** Subgroup analysis of UPDRS-III scores after the Baduanjin exercise, according to intervention duration.

Figure 5. a Total FMA-LE scores after the Baduanjin exercise. **b** Total FMA-LE scores after the Baduanjin exercise (sensitivity analysis).

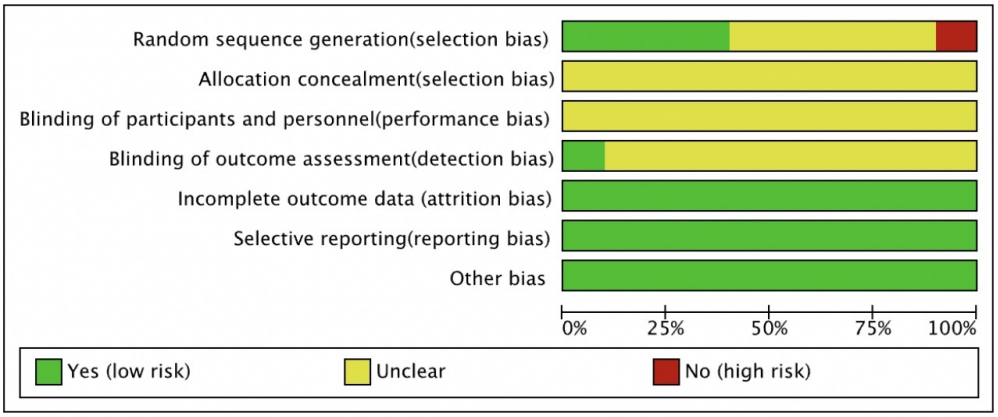
Figure 6. a Total BBS scores after the Baduanjin exercise. **b** Total BBS scores after the Baduanjin exercise (sensitivity analysis).

Figure 7. a Total 6WMD scores after the Baduanjin exercise. **b** Total 6WMD scores after the Baduanjin exercise (sensitivity analysis

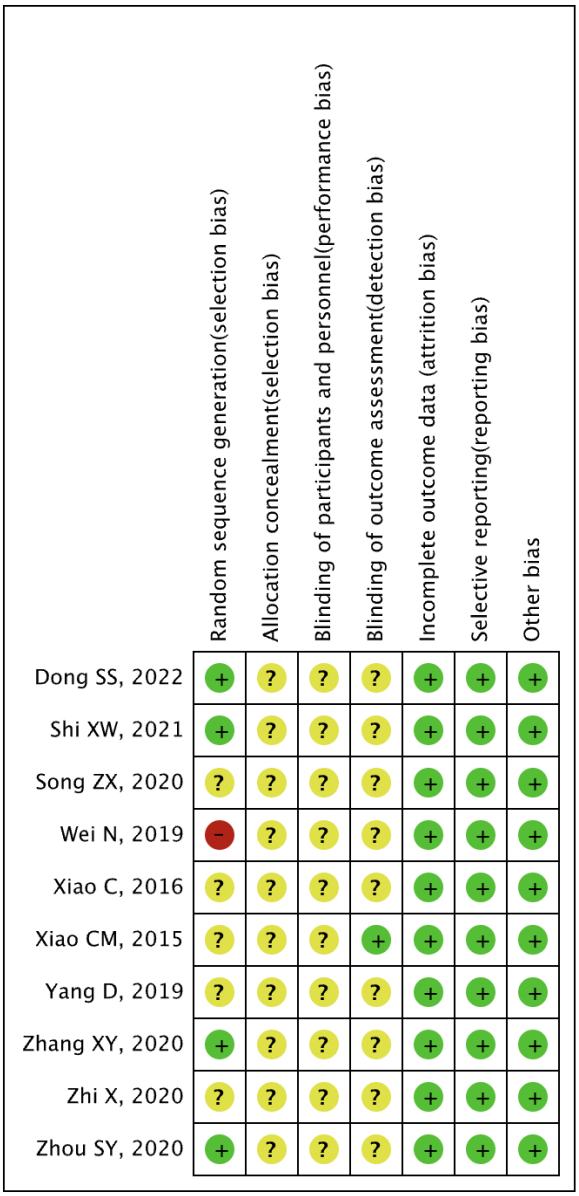
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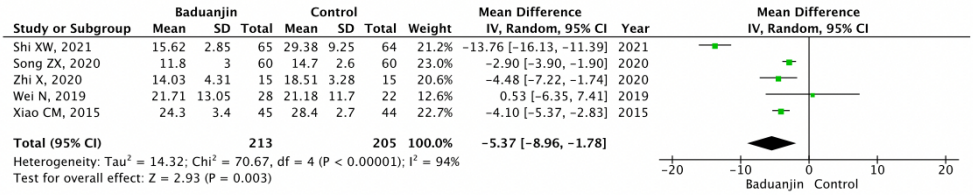
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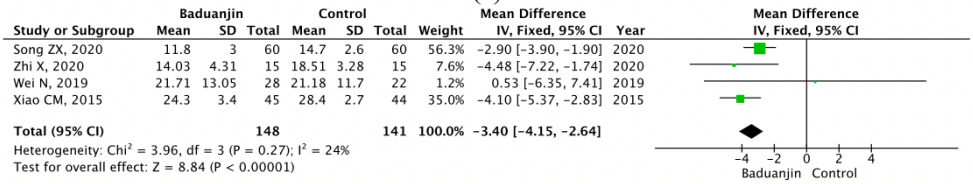
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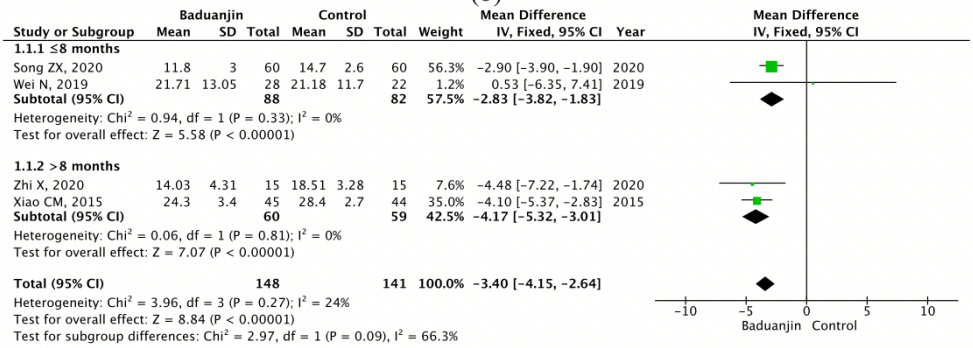
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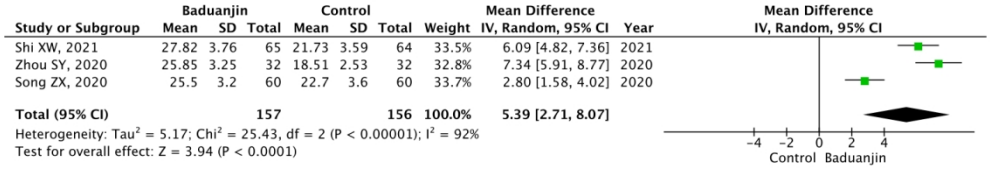


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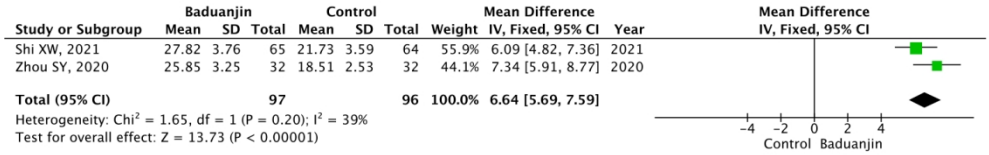


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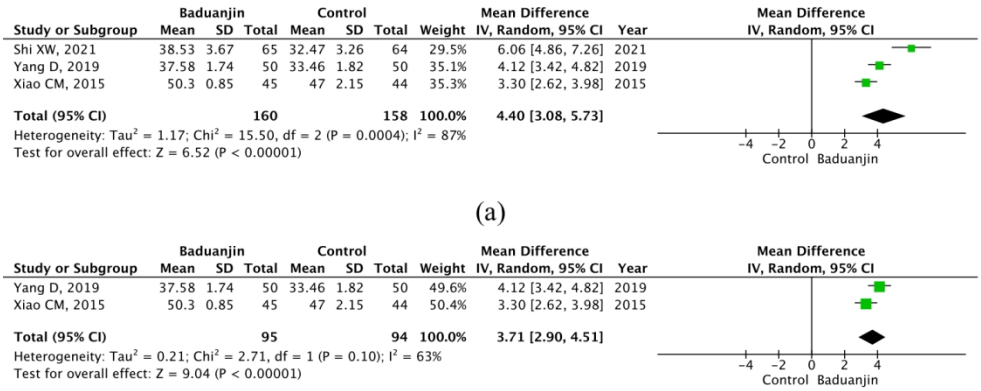


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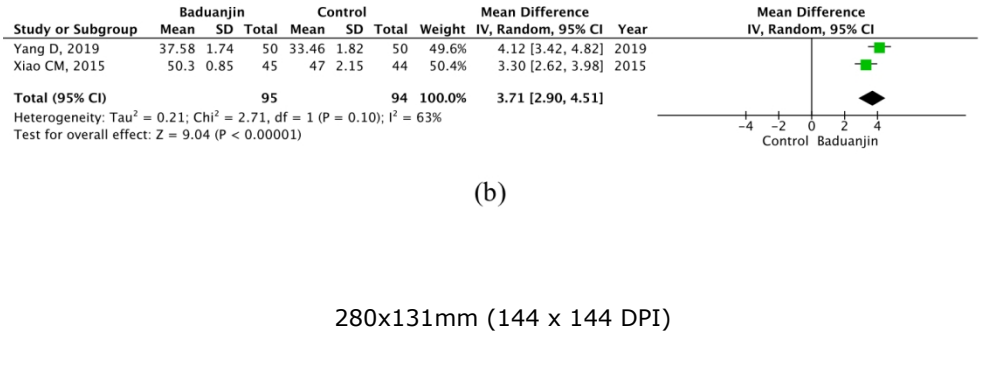


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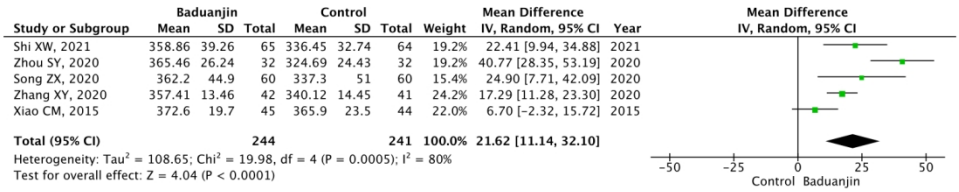


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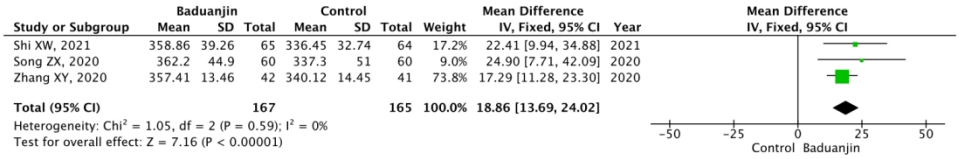


(b)

280x131mm (144 x 144 DPI)



(a)



(b)

287x145mm (144 x 144 DPI)

Search strategy

#1 Qigong[MeSH Terms]

#2 (((Baduanjin[Title/Abstract]) OR (Mind-body exercise[Title/Abstract])) OR (Ch'i Kung[Title/Abstract])) AND (traditional exercises[Title/Abstract])

#3 #1 OR #2

(Qigong[MeSH Terms]) OR ((((Baduanjin[Title/Abstract]) OR (Mind-body exercise[Title/Abstract])) OR (Ch'i Kung[Title/Abstract])) OR (traditional exercises[Title/Abstract]))

#4 (Parkinson disease[MeSH Terms]) OR (Parkinsonian Disorders[MeSH Terms])

#5 ((Parkinsonian Diseases[Title/Abstract]) OR (Parkinsonian Syndrome[Title/Abstract])) OR (PD[Title/Abstract])

#6 #4 OR #5

((Parkinson disease[MeSH Terms]) OR (Parkinsonian Disorders[MeSH Terms])) OR (((Parkinsonian Diseases[Title/Abstract]) OR (Parkinsonian Syndrome[Title/Abstract])) OR (PD[Title/Abstract]))

#7 #3 AND #6

((Qigong[MeSH Terms]) OR ((((Baduanjin[Title/Abstract]) OR (Mind-body exercise[Title/Abstract])) OR (Ch'i Kung[Title/Abstract])) OR (traditional exercises[Title/Abstract]))) AND (((Parkinson disease[MeSH Terms]) OR (Parkinsonian Disorders[MeSH Terms])) OR (((Parkinsonian Diseases[Title/Abstract]) OR (Parkinsonian Syndrome[Title/Abstract])) OR (PD[Title/Abstract])))

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page2
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page5
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page6
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page6
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	No
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page6
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page6
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	No
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page5

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	No
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Page7
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page9
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Figure 4-7
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page9-11
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Page9-11
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page9-11
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	No
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page9
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page9-11
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page11-13
	23b	Discuss any limitations of the evidence included in the review.	Page13
	23c	Discuss any limitations of the review processes used.	Page13
	23d	Discuss implications of the results for practice, policy, and future research.	Page13
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	No
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	No
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page14
Competing interests	26	Declare any competing interests of review authors.	Page14
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	No

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Effects of Baduanjin Exercise on Motor Function, Balance and Gait in Parkinson's Disease: A Systematic Review and Meta-analysis

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Manuscript ID	bmjopen-2022-067280.R1
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Date Submitted by the Author:	17-Oct-2022
Complete List of Authors:	Lai, Jinghui; Fujian University of Traditional Chinese Medicine Affiliated Rehabilitative Hospital, Five Departments of Encephalopathy Rehabilitation Cai, Yangfan; Fujian University of Traditional Chinese Medicine Affiliated Rehabilitative Hospital Yang, Liyan; Fujian University of Traditional Chinese Medicine Affiliated Rehabilitative Hospital Xia, Min; Fujian University of Traditional Chinese Medicine Affiliated Rehabilitative Hospital Cheng, Xi ; Fujian University of Traditional Chinese Medicine Affiliated Rehabilitative Hospital Chen, Ying; Fujian Agriculture and Forestry University
Primary Subject Heading:	Complementary medicine
Secondary Subject Heading:	Sports and exercise medicine, Rehabilitation medicine
Keywords:	Parkinson-s disease < NEUROLOGY, GERIATRIC MEDICINE, Health & safety < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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Effects of Baduanjin Exercise on Motor Function, Balance and Gait in Parkinson's Disease: A Systematic Review and Meta-analysis

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Abstract

Objective This study aims to systematically evaluate the effects of Baduanjin on motor function, balance and gait in Parkinson's disease (PD) patients.

Design Systematic review and meta-analysis.

Study selection All eligible randomized controlled trials (RCTs) published in the English and Chinese language were included.

Data sources Ten electronic databases were systematically searched, from inception to March 17, 2022: PubMed, Web of Science, Cochrane Library, Embase, EBSCOhost, OVID, SinoMed, China National Knowledge Infrastructure (CNKI), Wanfang Data, and China Science Journal Database (VIP).

Review methods Methodological quality assessment and meta-analysis were performed for the included studies using the Cochrane Review Manager Version 5.4 software.

Results Ten RCTs with 804 participants were included. The results revealed the following: (1) Baduanjin significantly improved the motor function of PD patients, based on the Unified Parkinson's Disease Rating Scale Part III (UPDRS-III; MD: -5.37, 95% CI: -8.96 to -1.78, P=0.003) and Fugl-Meyer Assessment of Lower Extremity (FMA-LE; MD: 5.39, 95% CI: 2.71 to 8.07, P<0.0001); (2) Baduanjin significantly improved the ability of balance of PD patients, based on the Berg Balance Scale (BBS; MD: 4.40, 95% CI: 3.08 to 5.73, P<0.00001); (3) Baduanjin significantly improved the gait of PD patients, based on the 6-minute walk distance (6MWD; MD: 21.62, 95% CI: 11.14 to 32.10, P<0.0001). After the further subgroup and sensitivity analyses, the heterogeneity was identified to be potentially due to the different degrees of disease severity in PD patients and the difference in Baduanjin intervention durations.

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Conclusions The analysis of this systematic evaluation indicates that Baduanjin might have a positive effect in improving the motor function, balance and gait of PD patients. However, due to the quantity and clinical heterogeneity limitations of the included studies, this conclusion still warrants more high-quality and multi-center RCTs for further verification.

Keywords Baduanjin; Parkinson's Disease; Motor Function; Balance; Gait; Meta-analysis

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Strengths and limitations of the study

- The study was the first to conduct a systematic review of the effectiveness and effects of Baduanjin on motor function, balance and gait in Parkinson's disease (PD) patients.
- Ten major databases were accessed for the study.
- There may be language bias, since merely studies in the English and Chinese language were retrieved.
- The bias of the original trials may have decreased the reliability of the results.

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Introduction

Parkinson's disease (PD) represents as the second most common neurodegenerative disease, which is characterized by the decrease of dopaminergic neurons in the substantia nigra pars compacta (SNpc).[1] According to the global survey of nervous system diseases, the rate of incidence and prevalence of PD have been dramatically increasing. In addition, this has been recognized as the potentially fastest-growing neurological disease in the world.[2,3] From 1990 to 2015, more than six million people suffered from PD.[4] Considering the increasing aging population, the number of PD patients is expected to reach more than 12 million by 2040, which would be almost double the present number.[5] PD cardinal clinical symptoms mainly include resting tremors, postural balance disorder, bradykinesia and rigidity.[6] Furthermore, PD patients are more prone to falls or fractures due to balance disorder and abnormal gait, limiting their activity ability, seriously affecting their quality of life (QoL), and imposing a heavy burden on families and society. Hence, the timely prevention or treatment of PD with effective strategies is imperative.

At present, levodopa remains as the main and first-line drug for the clinical treatment for PD patients.[7] Although drug therapy can improve the patient's symptoms, the effectiveness of pharmacological treatment tends to wear off over time, resulting in severe motor fluctuations and dyskinesia.[8] Furthermore, motor complications of any kind can severely impact motor control, and lead to increased disability and decreased QoL.[9] Given the high incidence of PD, the progressive nature of the disease, and the short duration of drug effectiveness, a pursuit for complementary and alternative medicine (CAM) would be essential to improve the QoL of patients, and minimize the burden on families and society.[10] The worldwide

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interest in using CAM to treat PD patients is growing. CAM include supplements, massage therapy, physical activity and mind-body exercises.[11]

Baduanjin is a traditional mind-body exercise that originated from the Song Dynasty, and has existed for more than one thousand years. This consists of eight simple movements, in which the whole set of activities are gentle, slow and continuous. This is characterized by deep and slow breathing, emphasizing static and dynamic posture control, and combining body movement and breathing patterns.[12] Baduanjin exercises can improve the patient's ability to focus on postural stability, strengthen core muscles, put pressure on weight-bearing joints, increase proprioception, and control body posture.[13] In addition, Baduanjin is commonly referred to as a moderate-intensity aerobic exercise that consumes 3-5 metabolic equivalents (MET), and is easy to learn and practice at home.[14] Based on its characteristics, this is very suitable for special populations, including populations with physical disabilities, such as patients with chronic diseases (*e.g.* PD), who have low tolerance to physical activity intensity.[15] In recent years, a number of trials have been conducted to investigate the effects of Baduanjin on PD. However, to date, no systematic evaluation of these research results has been identified. Therefore, in order to determine the effectiveness of Baduanjin in PD, a systematic review and meta-analysis of Baduanjin randomized controlled trials (RCTs) on PD patients was conducted, particularly in terms of the motor function, balance and gait of PD patients.

Methods

The present study was conducted in accordance with the guidelines of the Cochrane Handbook (PRISMA).[16]

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Patient and public involvement

No patient was involved in the study design.

Search strategy

An electronic literature search was performed on the following databases for articles published in the English and Chinese language as of March 2022: PubMed, Web of Science, Cochrane Library, Embase, EBSCOhost, OVID, SinoMed, China National Knowledge Infrastructure (CNKI), Wanfang Data, and China Science Journal Database (VIP). The following search terms were used: Baduanjin, Mind-body exercise, Qigong, and Parkinson's disease. All database search strategies are presented in the Appendix.

Inclusion and exclusion criteria

The inclusion criteria for the present study were, as follows: (1) studies that targeted PD patients; (2) RCTs; (3) the control group received other treatments (*e.g.* drug therapy, balance board training, conventional treatment, and conventional physical therapy), while the experimental group received Baduanjin exercises in addition to the interventions used in the control group; (4) for studies with multiple groups, all groups that met the above criteria were included; (5) outcome indicators: types correlated to motor function, balance and gait.

The exclusion criteria for the present study were, as follows: (1) the experimental group performed Baduanjin exercises combined with another intervention not used in the control group (*e.g.* Baduanjin combined with balance board training); (2)

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duplicate publications; (3) reviews, animal experiments, experience reports, *etc.*; (4) studies with no relevant results.

Study selection and data extraction

Two researchers independently screened the included studies and extracted the relevant data. The extracted data included the general information (first author and publication year, location, Hoehn and Yahr grade, age and sample size), details of the interventions (interventions measured in the control and Baduanjin groups, frequency of interventions, and intervention duration), and outcome indicators. Any disagreement during the process was resolved by discussion with a third researcher for confirmation.

Risk of bias

Two researchers used the bias risk assessment tool, which is applicable for RCTs and recommended by the Cochrane Manual 5.1.0, in order to independently evaluate the included literature and cross-check the results.[17] If there was any disagreement, a third researcher would arbitrate to reach a consensus. The risk of bias evaluation involved seven aspects: random sequence generation, allocation concealment, blinding of researchers and participants, blinding of evaluators, completeness of outcome indicators, selective reporting, and other bias.

Statistical analysis

The Cochrane Review Manager version 5.4 software was used for the data analysis. The count data were presented in odds ratio (OR) and 95% confidence intervals (CIs), and the measurement data was presented in mean difference (MD).

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The I^2-test was applied to estimate the statistical heterogeneity among trials. Studies with $I^2 \leq 50\%$ were considered to have low heterogeneity, and these were analyzed using a fixed effect model. Otherwise, significant heterogeneity was considered among the studies, and the random effects model was adopted. Then, a subgroup analysis (*e.g.* according to the duration of interventions) or sensitivity analysis was performed, and the possible reasons for the high heterogeneity were explored. Funnel plots were used to analyze the publication bias. The difference was defined to be statistically significant when $P<0.05$.

Results

Search results

A total of 267 articles were retrieved from the databases. Among these articles, 125 duplicate articles were excluded. Then, the titles and abstracts of the remaining 142 articles were screened, and 125 articles that were not qualified based on the criteria were excluded. After the full-text screening of the remaining 17 articles, a total of seven articles that did not meet the study criteria were excluded. The reasons for the exclusion were, as follows: non RCTs (n=4), clinical case reports (n=2), and duplicate publications (n=1). Finally, a total of 10 studies [18-27] were included for the present meta-analysis. The flowchart for the study selection is presented in Figure 1.

Characteristics of the included studies

The present review included a total of 10 RCTs with 804 participants diagnosed with PD.[18-27] For the language used by the included studies, eight [18-25] studies were reported in the Chinese language, and two [26-27] studies were reported in the English language. The Hoehn and Yahr grade is a standard clinical measure of PD progression,

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which globally measures the signs and symptoms of functional impairment. Grades 1-3 represents low-to-moderate incapacity, and grades 4-5 represents serious incapacity.[28] For the 10 included RCTs, five RCTs [18,20,23,24,27] clearly reported the disease grading of PD patients using the Hoehn and Yahr grade, while five RCTs [19,21,22,25,26] did not clearly report this. The control group received drug therapy, balance board training, conventional physical therapy, *etc.* The experimental group received any of the interventions in the control group plus Baduanjin. The Baduanjin intervention in the included studies had a frequency that ranged within 4-7 times a week, with a duration that lasted for 1-24 weeks. The main outcome indicators included the following: (1) tests of motor function, including the Unified Parkinson's Disease Rating Scale Part III (UPDRS-III), Fugl-Meyer Assessment of Lower Extremity (FMA-LE), and Function Independent Measure (FIM); (2) indicators of balance, including the Berg Balance Scale (BBS) and Balance-check® balancer test; (3) indicators of gait, including the 6-minute walk distance (6MWD), Timed Up and Go Test (TUGT), Intelligent Device for Energy Expenditure and Activity gait (IDEEA gait), Tinetti Performance Oriented Mobility Assessment (Tinetti POMA), 10-meter Walk Test (10MWT), and Freezing of Gait (FOG). The characteristics of the analyzed trials are presented in Table 1.

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Table 1. Basic characteristics of the included studies

Study	Location (language)	Hoehn & Yahr grades	Age	Sample (male/female)		Intervention		Frequency	Duration (weeks)	Intervention sites	Outcomes
				C	E	C	E				
Dong SS, 2022 [18]	Jiangsu, China (Chinese)	1-3	C: 63.07±12.78 E: 65.37 ± 7.47	30 (16/14)	27 (13/14)	Drug therapy	Drug therapy+Baduanjin	30 min/session, 1 session/day, 7 days/week	3	Hospital	IDEEA gait, Balance-check ® balancer test
Shi XW, 2021 [19]	Henan, China (Chinese)	Not reported	60-75	64 (34/30)	65 (34/31)	Drug therapy+Balance board training	Drug therapy+ Balance board training+Baduanjin	60 min/session, 1 session/day, 4 days/week	8	Hospital	FMA-LE, 6MWD, UPDRS-III, BBS
Song ZX, 2020 [20]	Jiangsu, China (Chinese)	1-3	>65	60 (36/24)	60 (32/28)	Drug therapy+Balance board training	Drug therapy+ Balance board training+Baduanjin	20 min/session, 1 session/day, 5 days/week	8	Hospital	FMA-LE, 6MWD, UPDRS-III, BBS
Zhang XY, 2020 [21]	Fuzhou, China (Chinese)	Not reported	61-74	41 (26/15)	42 (26/16)	Drug therapy+Balance board training	Drug therapy+ Balance board training+Baduanjin	30min/session, 1 session/day, 7 days/week	12	Hospital	6MWD, FIM
Zhou SY, 2020 [22]	Changsha, China (Chinese)	Not reported	60-86	39 (23/16)	39 (22/17)	Balance board training	Balance board training+Baduanjin	20 min/session, 1-2 sessions/day, 5 days/week	4	Hospital	FMA-LE, 6MWD
Zhi X, 2020 [23]	Dalian, China (Chinese)	1-2	45-65	15 (Not reported)	15 (Not reported)	Drug therapy	Drug therapy+ Baduanjin	60 min/session, 1 session/day, 6 days/week	12	Hospital	UPDRS-III, TUGT, 10MWT
Wei N, 2019 [24]	Beijing, China (Chinese)	1-3	32-78	22 (Not reported)	28 (Not reported)	Drug therapy+Health education	Drug therapy+Health education+Baduanjin	60 min/session, 1 session/day, 7 days/week	During hospitalization (1-2w)	Hospital	UPDRS-III
Yang D, 2019 [25]	Chongqing, China (Chinese)	Not reported	45-82	50 (31/19)	50 (30/20)	Conventional treatment	Conventional treatment+Baduanjin	Not reported	8	Hospital/ Home (after discharge)	BBS, Tinetti POMA

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Xiao C, 2016 [26]	Beijing, China (English)	Not reported	67.8 ± 9.4	33 (Not reported)	35 (Not reported)	Conventional physical therapy	Conventional physical therapy+Baduanjin	60 min/session, 1 session/day, 7 days/week	24	Not reported	BBS, TUGT, 6MWD, FOG, UPDRS-III
Xiao CM, 2015 [27]	Beijing, China (English)	1-3	55-80	44 (Not reported)	45 (Not reported)	Walking	Walking+Baduanjin	60 min/session, 1 session/day, 4 days/week	24	Home	UPDRS-III, BBS, TUGT, 6WMD

Note: C, control group; E, experiment group (Baduanjin group). **Outcomes:** IDEEA gait, Intelligent Device for Energy Expenditure and Activity gait; FMA-LE, Fugl-Meyer Assessment of Lower Extremity; 6MWD, 6-minute Walking Distance; FIM, Functional Independence Measurement; UPDRS-Ⅲ, Unified Parkinson's Disease Rating Scale, Part Ⅲ; BBS, Berg Balance Scale; TUGT, Timed Up and Go Test; 10MWT, 10-meter Walk Test; POMA, Performance Oriented Mobility Assessment; Tinetti POMA, Tinetti Performance Oriented Mobility Assessment; FOG, Freezing of Gait.

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Methodological quality and risk of bias

The methodological quality and risk of bias for each included study are presented in Figures 2 and 3. Four studies [18,19,21,22] reported the random sequence-generation method, while one study [27] used the blinded outcome assessors. All these studies were judged to have low risk of bias in the aforementioned domains. In addition, one study [24] was grouped according to the way patients voluntarily participated. Hence, this was rated as "high risk". All selective reporting and other bias risks included in the trial were considered as "low risk".

Meta-analysis

Effect of Baduanjin on motor function

Unified Parkinson's Disease Rating Scale Part III

UPDRS-III is part of the standard clinical test to assess the severity of motor complications in PD, with higher scores indicating more severe motor complications.[29] Five studies [19,20,23,24,27] with 418 participants used UPDRS-III to evaluate the motor symptoms. The meta-analysis revealed that the UPDRS-III scores were lower in the Baduanjin group, when compared to the control group (MD: -5.37, 95% CI: -8.96 to -1.78, P=0.003, I^2=94%; Fig. 4a). The heterogeneity sensitivity analysis was performed after excluding the study conducted by Shi *et al.* [19] The subsequent meta-analysis yielded the following results: MD: -3.40, 95% CI: -4.15 to -2.64, P<0.00001, I^2=24% (Fig. 4b). The heterogeneity may be due to the different degrees of PD among participants, and merely the study conducted by Shi *et al.* did not report the stages of the participant's PD. These above results indicate that Baduanjin can improve the degree of motor function impairment in PD.

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Furthermore, subgroup analyses were performed, according to the intervention durations. The meta-analysis of two studies [20,24] with a Baduanjin intervention duration of ≤8 weeks revealed the following: MD: -2.83, 95% CI: -3.82 to -1.83, P<0.00001, I^2=0%. The meta-analysis of two studies [23,27] with a Baduanjin intervention duration of >8 weeks revealed the following: MD: -4.17, 95% CI: -5.32 to -3.01, P<0.00001, I^2=0% (Fig. 4c). These results indicate that Baduanjin can reduce the degree of motor function impairment in PD, regardless of whether the intervention duration is ≤8 weeks or >8 weeks.

Fugl-Meyer Assessment of Lower Extremity

FMA-LE is a frequently used injury-based measure to assess the motor recovery in the lower extremity and predict the functional recovery of individuals with PD, with higher scores indicative of lower levels of impairment.[30] Three studies [19,20,22] that involved 313 participants used FMA-LE to assess the lower extremity motor function. The meta-analysis revealed that the FMA-LE scores were higher in the Baduanjin group, when compared to the control group (MD: 5.39, 95% CI: 2.71 to 8.07, P<0.0001, I^2=92%; Fig. 5a). Furthermore, a heterogeneity sensitivity analysis was performed after excluding the study conducted by Song *et al.* [20], and the subsequent meta-analysis results were consistent, with the following results: MD: 6.64, 95% CI: 5.69 to 7.59, P<0.00001, I^2=39% (Fig. 5b). The heterogeneity among these studies may be attributable to the difference in Baduanjin intervention duration. These results indicate that Baduanjin can improve the motor function of the lower limbs in PD patients.

Effect of Baduanjin on balance function

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Berg Balance Scale

BBS is a test used to assess functional balance, showing high validity and reliability in PD, with higher scores indicating better balance.[31] Three studies [19,25,27] that included 318 participants used BBS to evaluate the balance function. The meta-analysis revealed that the BBS scores were higher in the Baduanjin group, when compared to the control group (MD: 4.40, 95% CI: 3.08 to 5.73, P<0.00001, I^2=87%; Fig. 6a). A heterogeneity sensitivity analysis was performed after excluding the study conducted by Shi *et al.* [19], and the consistent results were, as follows: MD: 3.71, 95% CI: 2.90 to 4.51, P<0.00001, I^2=63% (Fig. 6b). These results indicate that Baduanjin has a positive effect on balance function in PD.

Effect of Baduanjin on gait

Six-minute walk distance

The 6MWD is a validated and widely used test to assess gait by instructing patients to walk for six minutes, and recording the longest distance the subject could walk.[32] Five studies [19-22,27] that involved 485 participants used the 6MWD to evaluate the walking ability. The results revealed that the 6MWD scores were higher in the Baduanjin group, when compared to the control group (MD: 21.62, 95% CI: 11.14 to 32.10, P<0.0001, I^2=80%; Fig. 7a). After removing the studies conducted by Zhou *et al.* [22] and Xiao *et al.* [27], the heterogeneity sensitivity analysis and subsequent meta-analysis revealed similar results: MD: 18.86, 95% CI: 13.69 to 24.02, P<0.00001, I^2=0% (Fig. 7b). The heterogeneity may have been attributed to the differences in duration of the Baduanjin interventions between the two studies [22,27] and other studies. These results indicate that Baduanjin can improve the gait of PD patients.

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Descriptive analysis

One study [26] presented the data in median (range). Hence, no combined analysis was conducted. The results of this study revealed that Baduanjin can improve the balance and lower limb strength of PD patients. Furthermore, two studies [23,26] used TUGT to evaluate the gait. However, due to the high heterogeneity and small number of pooled studies, the source of heterogeneity could not be identified. Hence, a meta-analysis was not conducted. One study [18] used IDEEA gait, and other study [25] used Tinetti POMA. In addition, one study [26] used FOG, and another one study [23] used the 10MWT. All these methods were used to assess the gait function, and the results suggest that Baduanjin can improve the gait function of PD patients. In addition, the study conducted by Zhang *et al.* [21] used FIM to assess the functional independence, and the results indicated that Baduanjin has a positive effect in improving the functional exercise capacity of PD patients. The study conducted by Dong *et al.* [18] used the Balance-check® balancer test to assess the balance, and the results revealed that Baduanjin can significantly improve the balance of PD patients.

Among the 10 included studies, no adverse events were reported, indicating that Baduanjin is a relatively safe exercise for PD patients. In order to ensure the compliance of patients with the intervention program, except for one study [26] that did not mention the specific measures, the remaining studies conducted centralized training through coaches during hospitalization or the supervision of family members at home after discharge, and the completion status was reported daily.

Publication bias

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According to the indicators of UPDRS-III, FMA-LE, BBS and 6MWD, the funnel chart was drawn, and the publication bias test was carried out. The results revealed that the figure for each indicator was not completely symmetrical between the left and right, suggesting that the studies may have a certain degree of publication bias (Fig. 8).

Discussion

The present study revealed that Baduanjin can reduce the UPDRS-III score, and increase the FMA-LE, BBS and 6MWD scores, indicating that Baduanjin might have a positive effect in improving the motor function, balance function, and gait of PD patients. In addition, the subgroup analysis for UPDRS-III revealed an improvement in motor function for PD patients, regardless of whether the duration of intervention was ≤8 weeks or >8 weeks.

Gaps in previous reviews

The present findings are contrary to Hao's review, which explored the effects of 10 different exercise interventions on motor function in PD, and revealed that Baduanjin has no benefit for PD patients. This may be correlated to the limited number of studies included in that study (merely three studies).[33] In addition, the conclusions of the present review were similar to those reported by another review, in which Baduanjin practice is beneficial for balance and trunk flexibility. However, that study included a population of healthy or people with different types of diseases.[34]

Effects of Baduanjin on motor function, balance function and gait

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As a chronic disease, PD has a long course, and is prone to recurrence, which is mainly caused by the progressive degeneration of dopaminergic neurons in the substantia nigra pars compacta, the decrease in striatal dopamine, and the presence of Lewy bodies.[35] Multiple abnormal neural circuits can lead to balance and gait problems in PD patients.[36] Gait problems are strongly dependent on impaired balance control, and the poor control of both can cause movement disorders.[37,38] Levodopa is the main drug for relieving motor symptoms of PD. However, approximately 50% of patients who receive levodopa for long durations would develop levodopa-induced dyskinesia (LID).[39] Speck *et al.* [40] suggested through a mice model that exercise can partially prevent the development of LID, and attenuate the side effects of levodopa through the normalization of striatopallidal dopaminergic signaling, without affecting the anti-parkinsonian effect of levodopa. Hence, this indicates that Baduanjin may induce an effect that enhances the medication efficiency and ameliorates the side effects of levodopa, making this a good CAM for the treatment of PD.

Several large controlled clinical studies have revealed that continuous exercise can improve the performance of daily activities in the early stage of PD, such as balance, gait and movement function, which is consistent with the present findings.[41,42] The exercise guidelines for PD recommends 20-60 minutes of moderate-intensity aerobic exercise for 3-5 days a week.[43] Baduanjin, as a moderate intensity aerobic exercise, mainly comprises of eight simple left-right symmetrical movements, such as arm raising, shaking of the head, waist rotation and foot flipping.[44] Practicing Baduanjin can fully stretch the joints of the upper and lower limbs, and the movement of the trunk can improve the ability of PD patients to control their center of gravity. At the same time, Baduanjin can also improve the

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proprioception, strengthen the coordination and sensitivity of limbs, and promote the recovery of balance function.[45] Ye *et al.* [46] reported that the upper limb exercise of Baduanjin can enhance the patient's coordination, and the knee flexion and squat movement can enhance the lower limb flexor and extensor muscle strength, and lower limb weight-bearing capacity, further improving the gait ability. This explains the findings reported in the present review.

The muscle tension caused by the uncoordinated contraction of active and antagonistic muscles influence a series of motor symptoms in patients with PD.[47] There is evidence that mind-body exercises, including Tai Chi, Yoga and Qigong, can reduce the contraction of antagonist muscles.[48,49] As a kind of mind-body exercise, the long-term practice of Baduanjin can promote blood circulation in the upper and lower limbs, and lower back, strengthened the muscle metabolism and stress function, improve the synergistic contraction of active muscles and antagonist muscles, and promote the recovery of motor function.[46] The α-syn protein is the main pathogenic protein of PD, and its accumulation is the main reason of neuronal loss.[50] Existing studies have revealed that aerobic exercise can reduce the loss of dopaminergic neurons, increase synaptic connections, and upregulate neurotrophic factor levels to improve PD dyskinesia. Furthermore, this can downregulate the α-Syn protein levels and neuronal apoptosis, which in turn, can reduce inflammation and mitochondrial dysfunction to restore motor function in PD patients.[51-53] Al-Jarrah *et al.* [54] proposed that aerobic exercise can decrease the level of proinflammatory proteins in the striatum in a PD experimental model by reducing the activity of microglia, thereby alleviating PD symptoms. These might be the possible mechanisms for the effect of Baduanjin in improving motor function.

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Limitations and future prospects

The present systematic review has several limitations. First, the present study included PD patients with Hoehn and Yahr grades of below grade 3 or no reported grade. Hence, the results cannot be extended to PD patients with Hoehn and Yahr grades 4-5. Second, the present study had a relatively high heterogeneity among the literatures, such as different study intervention durations, which might have been the cause for the clinical heterogeneity. Third, merely databases in the English and Chinese language were searched. Hence, it is possible that some researches may have been missed during the database search.

Conclusions

The present review revealed the potential benefits of Baduanjin in improving the motor function, balance function, and gait ability of PD patients. Therefore, Baduanjin exercise may be taken as a valuable aerobic exercise in clinical rehabilitation for the application of CAM. However, all findings must be interpreted with caution due to limitations in both the quantity and quality of available evidence. Furthermore, additional large, rigorous trials are warranted to better characterize the effects of Baduanjin in PD.

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Authors' contributions

Jinghui Lai, Yangfan Cai and Liyan Yang conceived and designed the study. Jinghui Lai drafted the manuscript. Yangfan Cai and Min Xia performed the data collection. Jinghui Lai, Xi Cheng and Ying Chen performed the data analysis. All authors approved the final manuscript before submission.

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Conflicts of interest

The authors declare that they have no conflict of interest.

Data sharing statement

All data relevant to the study are included in the article or uploaded as supplementary information.

Ethics approval

The study did not involve human participants.

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Figure legends

Figure 1 Literature screening process.

Figure 2 Risk of bias graph.

Figure 3 Risk of bias summary.

Figure 4 (a) Total UPDRS-III scores after the Baduanjin exercise. (b) Total UPDRS-III scores after the Baduanjin exercise (sensitivity analysis). (c) Subgroup analysis of the UPDRS-III scores after the Baduanjin exercise, according to intervention duration.

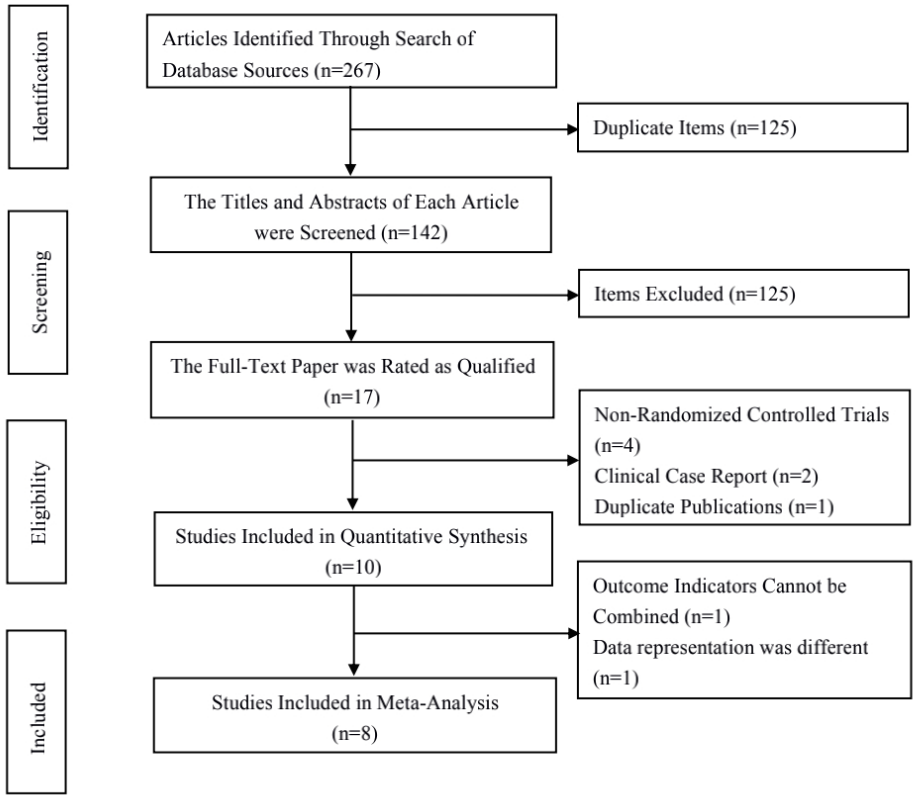
Figure 5 (a) Total FMA-LE scores after the Baduanjin exercise. (b) Total FMA-LE scores after the Baduanjin exercise (sensitivity analysis).

Figure 6 (a) Total BBS scores after the Baduanjin exercise. (b) Total BBS scores after the Baduanjin exercise (sensitivity analysis).

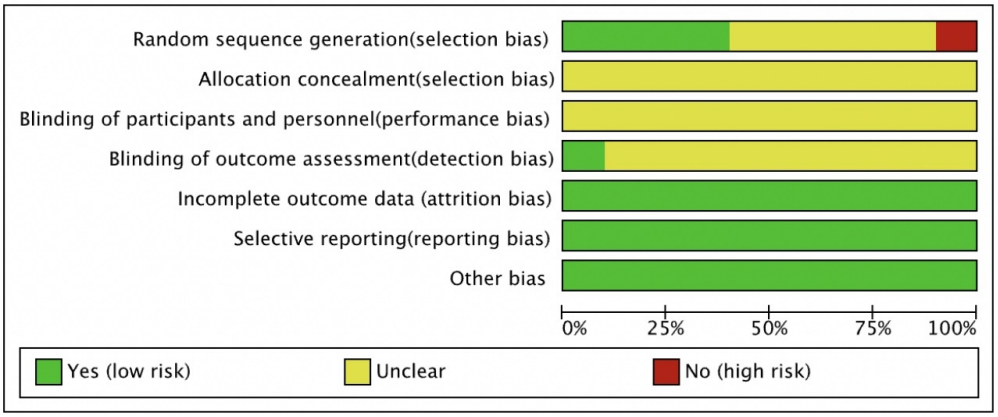
Figure 7 (a) Total 6WMD scores after the Baduanjin exercise. (b) Total 6WMD scores after the Baduanjin exercise (sensitivity analysis).

Figure 8 (a) Funnel chart for UPDRS-III. (b) Funnel chart for FMA-LE. (c) Funnel chart for BBS. (d) Funnel chart for 6MWD.

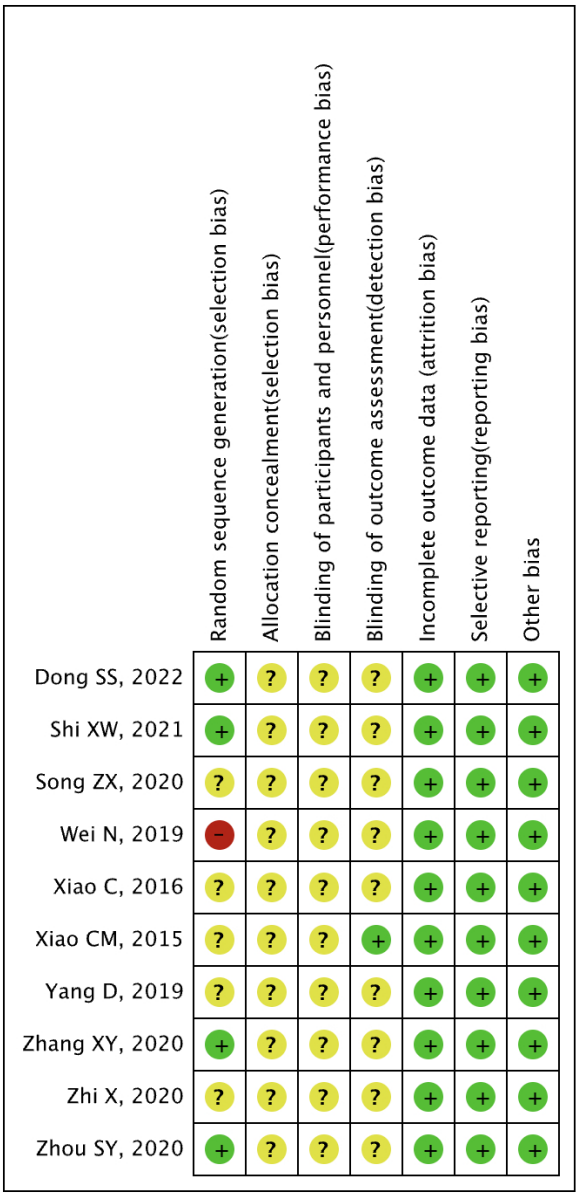
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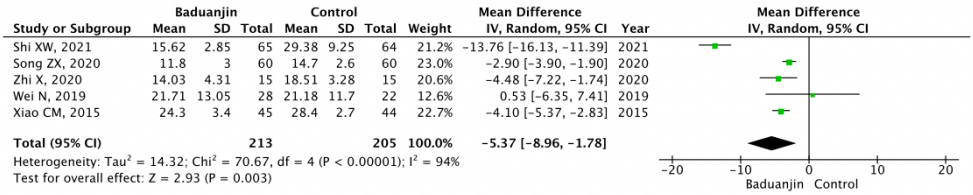
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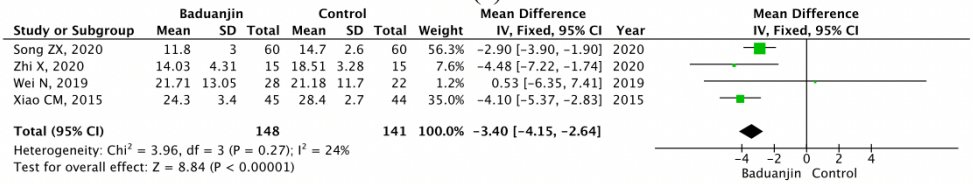
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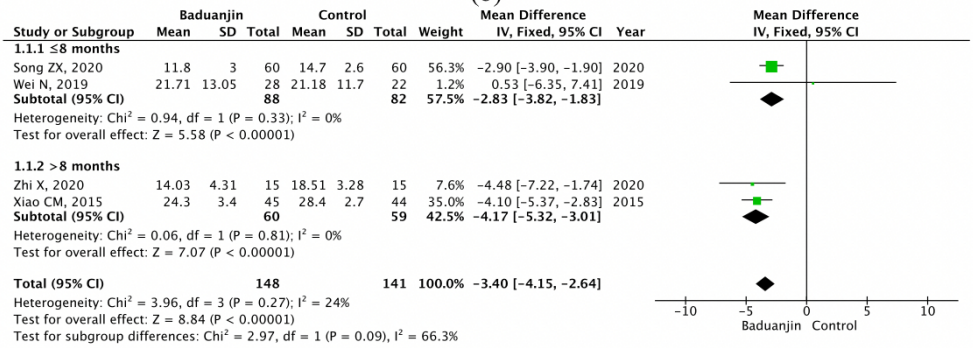
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(a)

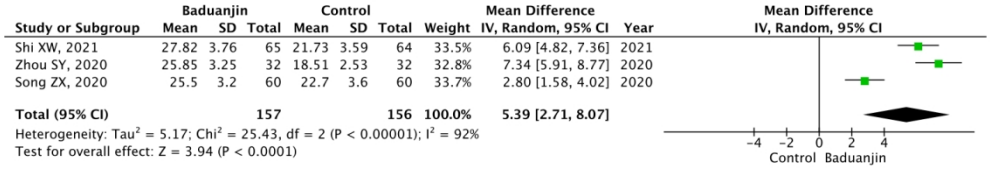


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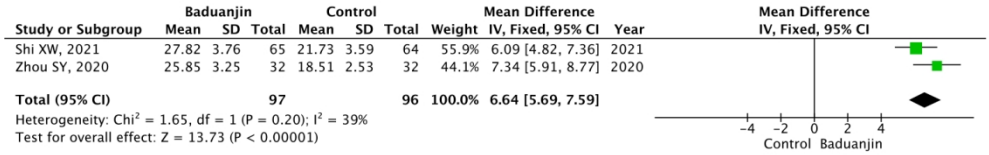


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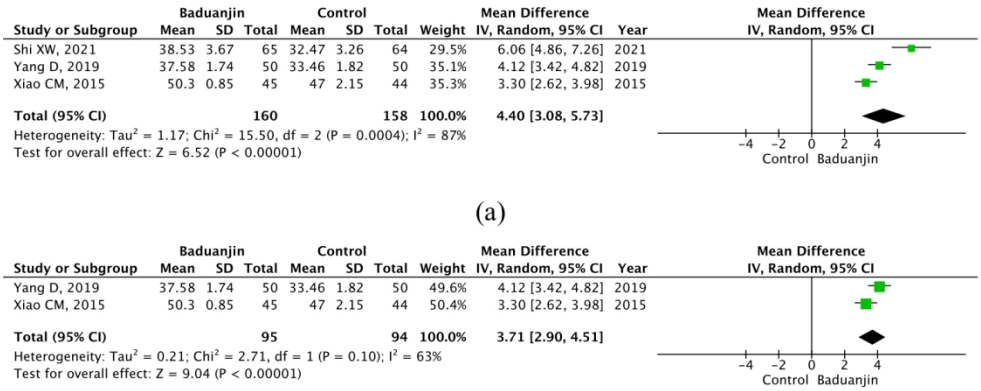


(a)



(b)

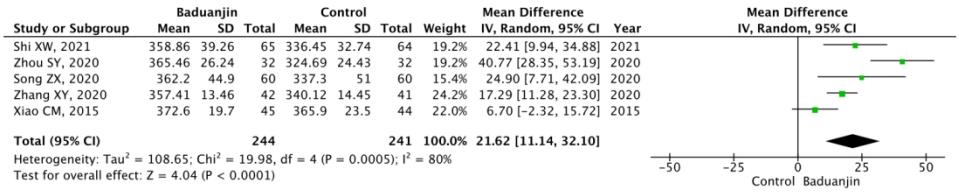
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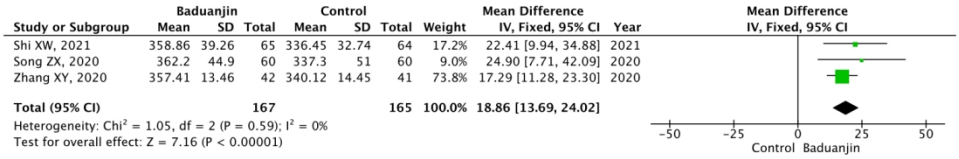
(a)

(b)

280x131mm (144 x 144 DPI)

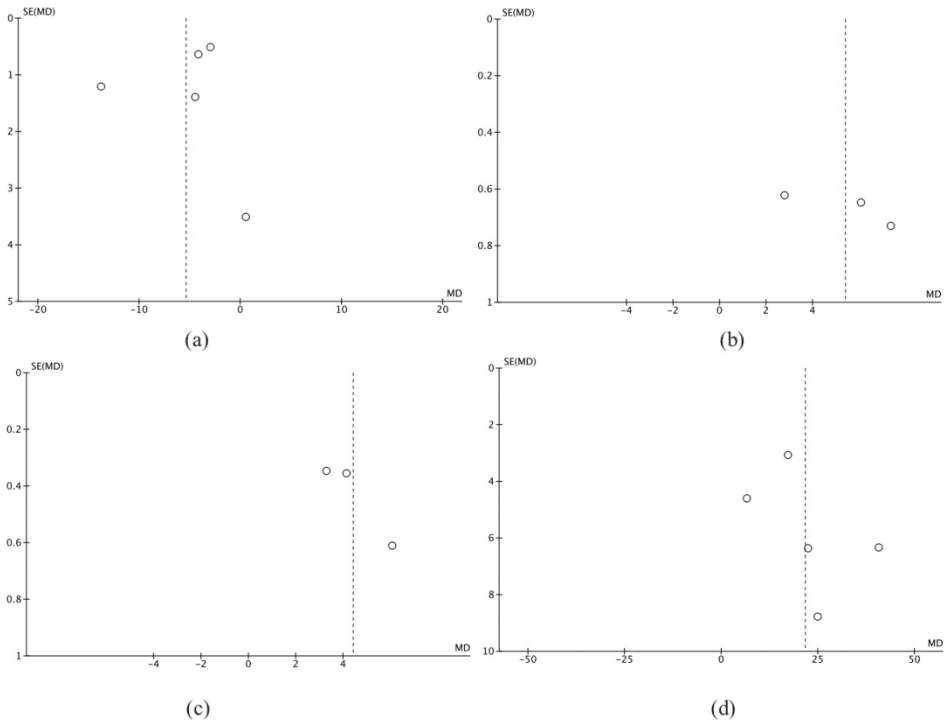


(a)



(b)

287x145mm (144 x 144 DPI)



277x201mm (144 x 144 DPI)

Search strategy

PubMed

#1 Qigong[MeSH Terms] Filters: Chinese, English.

#2 (((Baduanjin[Title/Abstract]) OR (Mind-body exercise[Title/Abstract])) OR (Ch'i Kung[Title/Abstract])) AND (traditional exercises[Title/Abstract]) Filters: Chinese, English.

#3 #1 OR #2

#4 (Parkinson disease[MeSH Terms]) OR (Parkinsonian Disorders[MeSH Terms]) Filters: Chinese, English.

#5 ((Parkinsonian Diseases[Title/Abstract]) OR (Parkinsonian Syndrome[Title/Abstract])) OR (PD[Title/Abstract]) Filters: Chinese, English.

#6 #4 OR #5

#7 #3 AND #6

((Qigong[MeSH Terms]) OR ((((Baduanjin[Title/Abstract]) OR (Mind-body exercise[Title/Abstract])) OR (Ch'i Kung[Title/Abstract])) OR (traditional exercises[Title/Abstract]))) AND (((Parkinson disease[MeSH Terms]) OR (Parkinsonian Disorders[MeSH Terms])) OR (((Parkinsonian Diseases[Title/Abstract]) OR (Parkinsonian Syndrome[Title/Abstract])) OR (PD[Title/Abstract]))) Filters: Chinese, English.

Web of Science

#1 ((((TS=(Qigong)) OR TS=(Baduanjin)) OR TS=(Mind-body exercise)) OR TS=(Ch'i Kung)) OR TS=(traditional exercises)

#2 ((((TS=(Parkinson disease)) OR TS=(Parkinsonian Disorders)) OR TS=(Parkinsonian Diseases)) OR TS=(Parkinsonian Syndrome)) OR TS=(PD)

#3 #1 AND #2

((((TS=(Qigong)) OR TS=(Baduanjin)) OR TS=(Mind-body exercise)) OR TS=(Ch'i Kung)) OR TS=(traditional exercises) AND ((((TS=(Parkinson disease)) OR TS=(Parkinsonian Disorders)) OR TS=(Parkinsonian Diseases)) OR TS=(Parkinsonian Syndrome)) OR TS=(PD) AND English OR Chinese (Languages)

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Cochrane Library

#1 MeSH descriptor: [Qigong] explode all trees

#2 (Baduanjin OR Mind-body exercise OR Ch'i Kung OR traditional exercises):ti,ab,kw

#3 #1 OR #2

#4 MeSH descriptor: [Parkinson Disease] explode all trees

#5 MeSH descriptor: [Parkinsonian Disorders] explode all trees

#6 (Parkinsonian Diseases OR Parkinsonian Syndrome OR PD):ti,ab,kw

#7 #4 OR #5 OR #6

#8 #3 AND #7

Embase

#1 'qigong'/exp AND ([chinese]/lim OR [english]/lim)

#2 ('baduanjin':ab,ti OR 'mind-body exercise':ab,ti OR 'chi kung':ab,ti OR 'traditional exercises':ab,ti) AND ([chinese]/lim OR [english]/lim)

#3 #1 OR #2

#4 'parkinson disease'/exp OR 'parkinsonian disorders'/exp AND ([chinese]/lim OR [english]/lim)

#5 ('parkinsonian diseases':ab,ti OR 'parkinsonian syndrome':ab,ti OR pd:ab,ti) AND ([chinese]/lim OR [english]/lim)

#6 #4 OR #5

#7 #3 AND #6

EBSCOhost

#1 SU qigong Language:- english or chinese

#2 TX (Baduanjin OR Mind-body exercise OR Ch'i Kung OR traditional exercises) Language:- english or chinese

#3 #1 OR #2 SU qigong OR TX (Baduanjin OR Mind-body exercise OR Ch'i Kung OR traditional exercises)

#4 SU (parkinson disease OR parkinsonian disorders) Language:- english or chinese

#5 TX (Parkinsonian Diseases OR Parkinsonian Syndrome OR PD) Language:- english or chinese

#6 #4 OR #5 SU (parkinson disease OR parkinsonian disorders) OR (Parkinsonian Diseases OR

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Parkinsonian Syndrome OR PD)

#7 #3 AND #6

OVID

#1 qigong.hw. limit (chinese or english)

#2 (Baduanjin or Mind-body exercise or Ch'i Kung or traditional exercises).af. limit (chinese or english)

#3 #1 or #2 qigong.hw. or (Baduanjin or Mind-body exercise or Ch'i Kung or traditional exercises).af.

#4 (parkinson disease or parkinsonian disorders).hw. limit (chinese or english)

#5 (Parkinsonian Diseases or Parkinsonian Syndrome or PD).af. limit (chinese or english)

#6 #4 or #5 (parkinson disease or parkinsonian disorders) or (Parkinsonian Diseases or Parkinsonian Syndrome or PD).af.

#7 #3 and #6

SinoMed

#1 ("气功"[加权:扩展])

#2 "八段锦"[中文标题:智能] OR "传统功法"[中文标题:智能] OR "身心运动"[中文标题:智能] OR "体育疗法"[中文标题:智能]

#3 "八段锦"[摘要:智能] OR "传统功法"[摘要:智能] OR "身心运动"[摘要:智能] OR "体育疗法"[摘要:智能]

#4 #1 OR #2 OR #3

#5 "帕金森病"[加权:扩展]

#6 "帕金森综合征"[中文标题:智能] OR "帕金森障碍"[中文标题:智能]

#7 "帕金森综合征"[摘要:智能] OR "帕金森障碍"[摘要:智能]

#8 #5 OR #6 OR #7

#9 #4 AND #8

China National Knowledge Infrastructure (CNKI)

#1 SU=八段锦+传统功法+身心运动+气功+体育疗法

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#2 SU=帕金森病+帕金森综合征+帕金森障碍

#3 #1 and #2

(SU=八段锦+传统功法+身心运动+气功+体育疗法) and (SU=帕金森病+帕金森综合征+帕金森障碍)

Wanfang Data

#1 主题:八段锦+传统功法+身心运动+气功+体育疗法

#2 主题:帕金森病+帕金森综合征+帕金森障碍

#3 #1 and #2

(主题:八段锦+传统功法+身心运动+气功+体育疗法) and (主题:帕金森病+帕金森综合征+帕金森障碍)

China Science Journal Database (VIP)

#1 M=八段锦 OR M=传统功法 OR M=身心运动 OR M=气功 OR M=体育疗法

#2 M=帕金森病 OR M=帕金森综合征 OR M=帕金森障碍

#3 #1 AND #2

(M=八段锦 OR M=传统功法 OR M=身心运动 OR M=气功 OR M=体育疗法) AND (M=帕金森病 OR M=帕金森综合征 OR M=帕金森障碍)

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page2
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page5
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page6
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page6
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	No
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page6
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page6
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	No
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page5

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	No
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Page7
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page9
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Figure 4-7
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page9-11
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Page9-11
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page9-11
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	No
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page9
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page9-11
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page11-13
	23b	Discuss any limitations of the evidence included in the review.	Page13
	23c	Discuss any limitations of the review processes used.	Page13
	23d	Discuss implications of the results for practice, policy, and future research.	Page13
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	No
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	No
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page14
Competing interests	26	Declare any competing interests of review authors.	Page14
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	No

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Effects of Baduanjin Exercise on Motor Function, Balance and Gait in Parkinson's Disease: A Systematic Review and Meta-analysis

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Manuscript ID	bmjopen-2022-067280.R2
Article Type:	Original research
Date Submitted by the Author:	24-Oct-2022
Complete List of Authors:	Lai, Jinghui; Fujian University of Traditional Chinese Medicine Affiliated Rehabilitative Hospital, Five Departments of Encephalopathy Rehabilitation Cai, Yangfan; Fujian University of Traditional Chinese Medicine Affiliated Rehabilitative Hospital Yang, Liyan; Fujian University of Traditional Chinese Medicine Affiliated Rehabilitative Hospital Xia, Min; Fujian University of Traditional Chinese Medicine Affiliated Rehabilitative Hospital Cheng, Xi ; Fujian University of Traditional Chinese Medicine Affiliated Rehabilitative Hospital Chen, Ying; Fujian Agriculture and Forestry University
Primary Subject Heading:	Complementary medicine
Secondary Subject Heading:	Sports and exercise medicine, Rehabilitation medicine
Keywords:	Parkinson-s disease < NEUROLOGY, GERIATRIC MEDICINE, Health & safety < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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Effects of Baduanjin Exercise on Motor Function, Balance and Gait in Parkinson's Disease: A Systematic Review and Meta-analysis

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Abstract

Objective This study aims to systematically evaluate the effects of Baduanjin on motor function, balance and gait in Parkinson's disease (PD) patients.

Design Systematic review and meta-analysis.

Study selection All eligible randomized controlled trials (RCTs) published in the English and Chinese language were included.

Data sources Ten electronic databases were systematically searched, from inception to March 17, 2022: PubMed, Web of Science, Cochrane Library, Embase, EBSCOhost, OVID, SinoMed, China National Knowledge Infrastructure (CNKI), Wanfang Data, and China Science Journal Database (VIP).

Review methods Methodological quality assessment and meta-analysis were performed for the included studies using the Cochrane Review Manager Version 5.4 software.

Results Ten RCTs with 804 participants were included. The results revealed the following: (1) Baduanjin significantly improved the motor function of PD patients, based on the Unified Parkinson's Disease Rating Scale Part III (UPDRS-III; MD: -5.37, 95% CI: -8.96 to -1.78, P=0.003) and Fugl-Meyer Assessment of Lower Extremity (FMA-LE; MD: 5.39, 95% CI: 2.71 to 8.07, P<0.0001); (2) Baduanjin significantly improved the ability of balance of PD patients, based on the Berg Balance Scale (BBS; MD: 4.40, 95% CI: 3.08 to 5.73, P<0.00001); (3) Baduanjin significantly improved the gait of PD patients, based on the 6-minute walk distance (6MWD; MD: 21.62, 95% CI: 11.14 to 32.10, P<0.0001). After the further subgroup and sensitivity analyses, the heterogeneity was identified to be potentially due to the different degrees of disease severity in PD patients and the difference in Baduanjin intervention durations.

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Conclusions The analysis of this systematic evaluation indicates that Baduanjin might have a positive effect in improving the motor function, balance and gait of PD patients. However, due to the quantity and clinical heterogeneity limitations of the included studies, this conclusion still warrants more high-quality and multi-center RCTs for further verification.

Keywords Baduanjin; Parkinson's Disease; Motor Function; Balance; Gait; Meta-analysis

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Strengths and limitations of the study

- The study was the first to conduct a systematic review of the effectiveness and effects of Baduanjin on motor function, balance and gait in Parkinson's disease (PD) patients.
- Ten major databases were accessed for the study.
- There may be language bias, since merely studies in the English and Chinese language were retrieved.
- The bias of the original trials may have decreased the reliability of the results.

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Introduction

Parkinson's disease (PD) represents as the second most common neurodegenerative disease, which is characterized by the decrease of dopaminergic neurons in the substantia nigra pars compacta (SNpc).[1] According to the global survey of nervous system diseases, the rate of incidence and prevalence of PD have been dramatically increasing. In addition, this has been recognized as the potentially fastest-growing neurological disease in the world.[2,3] From 1990 to 2015, more than six million people suffered from PD.[4] Considering the increasing aging population, the number of PD patients is expected to reach more than 12 million by 2040, which would be almost double the present number.[5] PD cardinal clinical symptoms mainly include resting tremors, postural balance disorder, bradykinesia and rigidity.[6] Furthermore, PD patients are more prone to falls or fractures due to balance disorder and abnormal gait, limiting their activity ability, seriously affecting their quality of life (QoL), and imposing a heavy burden on families and society. Hence, the timely prevention or treatment of PD with effective strategies is imperative.

At present, levodopa remains as the main and first-line drug for the clinical treatment for PD patients.[7] Although drug therapy can improve the patient's symptoms, the effectiveness of pharmacological treatment tends to wear off over time, resulting in severe motor fluctuations and dyskinesia.[8] Furthermore, motor complications of any kind can severely impact motor control, and lead to increased disability and decreased QoL.[9] Given the high incidence of PD, the progressive nature of the disease, and the short duration of drug effectiveness, a pursuit for complementary and alternative medicine (CAM) would be essential to improve the QoL of patients, and minimize the burden on families and society.[10] The worldwide

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interest in using CAM to treat PD patients is growing. CAM include supplements, massage therapy, physical activity and mind-body exercises.[11]

Baduanjin is a traditional mind-body exercise that originated from the Song Dynasty, and has existed for more than one thousand years. This consists of eight simple movements, in which the whole set of activities are gentle, slow and continuous. This is characterized by deep and slow breathing, emphasizing static and dynamic posture control, and combining body movement and breathing patterns.[12] Baduanjin exercises can improve the patient's ability to focus on postural stability, strengthen core muscles, put pressure on weight-bearing joints, increase proprioception, and control body posture.[13] In addition, Baduanjin is commonly referred to as a moderate-intensity aerobic exercise that consumes 3-5 metabolic equivalents (MET), and is easy to learn and practice at home.[14] Based on its characteristics, this is very suitable for special populations, including populations with physical disabilities, such as patients with chronic diseases (*e.g.* PD), who have low tolerance to physical activity intensity.[15] In recent years, a number of trials have been conducted to investigate the effects of Baduanjin on PD. However, to date, no systematic evaluation of these research results has been identified. Therefore, in order to determine the effectiveness of Baduanjin in PD, a systematic review and meta-analysis of Baduanjin randomized controlled trials (RCTs) on PD patients was conducted, particularly in terms of the motor function, balance and gait of PD patients.

Methods

The present study was conducted in accordance with the guidelines of the Cochrane Handbook (PRISMA).[16]

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Patient and public involvement

No patient was involved in the study design.

Search strategy

An electronic literature search was performed on the following databases for articles published in the English and Chinese language as of March 2022: PubMed, Web of Science, Cochrane Library, Embase, EBSCOhost, OVID, SinoMed, China National Knowledge Infrastructure (CNKI), Wanfang Data, and China Science Journal Database (VIP). The following search terms were used: Baduanjin, Mind-body exercise, Qigong, and Parkinson's disease. All database search strategies are presented in the Appendix. (See Appendix 1)

Inclusion and exclusion criteria

The inclusion criteria for the present study were, as follows: (1) studies that targeted PD patients; (2) RCTs; (3) the control group received other treatments (*e.g.* drug therapy, balance board training, conventional treatment, and conventional physical therapy), while the experimental group received Baduanjin exercises in addition to the interventions used in the control group; (4) for studies with multiple groups, all groups that met the above criteria were included; (5) outcome indicators: types correlated to motor function, balance and gait.

The exclusion criteria for the present study were, as follows: (1) the experimental group performed Baduanjin exercises combined with another intervention not used in the control group (*e.g.* Baduanjin combined with balance board training); (2)

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duplicate publications; (3) reviews, animal experiments, experience reports, *etc.*; (4) studies with no relevant results.

Study selection and data extraction

Two researchers independently screened the included studies and extracted the relevant data. The extracted data included the general information (first author and publication year, location, Hoehn and Yahr grade, age and sample size), details of the interventions (interventions measured in the control and Baduanjin groups, frequency of interventions, and intervention duration), and outcome indicators. Any disagreement during the process was resolved by discussion with a third researcher for confirmation.

Risk of bias

Two researchers used the bias risk assessment tool, which is applicable for RCTs and recommended by the Cochrane Manual 5.1.0, in order to independently evaluate the included literature and cross-check the results.[17] If there was any disagreement, a third researcher would arbitrate to reach a consensus. The risk of bias evaluation involved seven aspects: random sequence generation, allocation concealment, blinding of researchers and participants, blinding of evaluators, completeness of outcome indicators, selective reporting, and other bias.

Statistical analysis

The Cochrane Review Manager version 5.4 software was used for the data analysis. The count data were presented in odds ratio (OR) and 95% confidence intervals (CIs), and the measurement data was presented in mean difference (MD).

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The I^2-test was applied to estimate the statistical heterogeneity among trials. Studies with $I^2 \leq 50\%$ were considered to have low heterogeneity, and these were analyzed using a fixed effect model. Otherwise, significant heterogeneity was considered among the studies, and the random effects model was adopted. Then, a subgroup analysis (*e.g.* according to the duration of interventions) or sensitivity analysis was performed, and the possible reasons for the high heterogeneity were explored. Funnel plots were used to analyze the publication bias. The difference was defined to be statistically significant when $P<0.05$.

Results

Search results

A total of 267 articles were retrieved from the databases. Among these articles, 125 duplicate articles were excluded. Then, the titles and abstracts of the remaining 142 articles were screened, and 125 articles that were not qualified based on the criteria were excluded. After the full-text screening of the remaining 17 articles, a total of seven articles that did not meet the study criteria were excluded. The reasons for the exclusion were, as follows: non RCTs (n=4), clinical case reports (n=2), and duplicate publications (n=1). Finally, a total of 10 studies [18-27] were included for the present meta-analysis. The flowchart for the study selection is presented in Figure 1.

Characteristics of the included studies

The present review included a total of 10 RCTs with 804 participants diagnosed with PD.[18-27] For the language used by the included studies, eight [18-25] studies were reported in the Chinese language, and two [26-27] studies were reported in the English language. The Hoehn and Yahr grade is a standard clinical measure of PD progression,

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which globally measures the signs and symptoms of functional impairment. Grades 1-3 represents low-to-moderate incapacity, and grades 4-5 represents serious incapacity.[28] For the 10 included RCTs, five RCTs [18,20,23,24,27] clearly reported the disease grading of PD patients using the Hoehn and Yahr grade, while five RCTs [19,21,22,25,26] did not clearly report this. The control group received drug therapy, balance board training, conventional physical therapy, *etc.* The experimental group received any of the interventions in the control group plus Baduanjin. The Baduanjin intervention in the included studies had a frequency that ranged within 4-7 times a week, with a duration that lasted for 1-24 weeks. The main outcome indicators included the following: (1) tests of motor function, including the Unified Parkinson's Disease Rating Scale Part III (UPDRS-III), Fugl-Meyer Assessment of Lower Extremity (FMA-LE), and Function Independent Measure (FIM); (2) indicators of balance, including the Berg Balance Scale (BBS) and Balance-check® balancer test; (3) indicators of gait, including the 6-minute walk distance (6MWD), Timed Up and Go Test (TUGT), Intelligent Device for Energy Expenditure and Activity gait (IDEEA gait), Tinetti Performance Oriented Mobility Assessment (Tinetti POMA), 10-meter Walk Test (10MWT), and Freezing of Gait (FOG). The characteristics of the analyzed trials are presented in Table 1.

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Table 1. Basic characteristics of the included studies

Study	Location (language)	Hoehn & Yahr grades	Age	Sample (male/female)		Intervention		Frequency	Duration (weeks)	Intervention sites	Outcomes
				C	E	C	E				
Dong SS, 2022 [18]	Jiangsu, China (Chinese)	1-3	C: 63.07±12.78 E: 65.37 ± 7.47	30 (16/14)	27 (13/14)	Drug therapy	Drug therapy+Baduanjin	30 min/session, 1 session/day, 7 days/week	3	Hospital	IDEEA gait, Balance-check ® balancer test
Shi XW, 2021 [19]	Henan, China (Chinese)	Not reported	60-75	64 (34/30)	65 (34/31)	Drug therapy+Balance board training	Drug therapy+ Balance board training+Baduanjin	60 min/session, 1 session/day, 4 days/week	8	Hospital	FMA-LE, 6MWD, UPDRS-III, BBS
Song ZX, 2020 [20]	Jiangsu, China (Chinese)	1-3	>65	60 (36/24)	60 (32/28)	Drug therapy+Balance board training	Drug therapy+ Balance board training+Baduanjin	20 min/session, 1 session/day, 5 days/week	8	Hospital	FMA-LE, 6MWD, UPDRS-III, BBS
Zhang XY, 2020 [21]	Fuzhou, China (Chinese)	Not reported	61-74	41 (26/15)	42 (26/16)	Drug therapy+Balance board training	Drug therapy+ Balance board training+Baduanjin	30min/session, 1 session/day, 7 days/week	12	Hospital	6MWD, FIM
Zhou SY, 2020 [22]	Changsha, China (Chinese)	Not reported	60-86	39 (23/16)	39 (22/17)	Balance board training	Balance board training+Baduanjin	20 min/session, 1-2 sessions/day, 5 days/week	4	Hospital	FMA-LE, 6MWD
Zhi X, 2020 [23]	Dalian, China (Chinese)	1-2	45-65	15 (Not reported)	15 (Not reported)	Drug therapy	Drug therapy+ Baduanjin	60 min/session, 1 session/day, 6 days/week	12	Hospital	UPDRS-III, TUGT, 10MWT
Wei N, 2019 [24]	Beijing, China (Chinese)	1-3	32-78	22 (Not reported)	28 (Not reported)	Drug therapy+Health education	Drug therapy+Health education+Baduanjin	60 min/session, 1 session/day, 7 days/week	During hospitalization (1-2w)	Hospital	UPDRS-III
Yang D, 2019 [25]	Chongqing, China (Chinese)	Not reported	45-82	50 (31/19)	50 (30/20)	Conventional treatment	Conventional treatment+Baduanjin	Not reported	8	Hospital/ Home (after discharge)	BBS, Tinetti POMA

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Xiao C, 2016 [26]	Beijing, China (English)	Not reported	67.8 ± 9.4	33 (Not reported)	35 (Not reported)	Conventional physical therapy	Conventional physical therapy+Baduanjin	60 min/session, 1 session/day, 7 days/week	24	Not reported	BBS, TUGT, 6MWD, FOG, UPDRS-III
Xiao CM, 2015 [27]	Beijing, China (English)	1-3	55-80	44 (Not reported)	45 (Not reported)	Walking	Walking+Baduanjin	60 min/session, 1 session/day, 4 days/week	24	Home	UPDRS-III, BBS, TUGT, 6WMD

Note: C, control group; E, experiment group (Baduanjin group). **Outcomes:** IDEEA gait, Intelligent Device for Energy Expenditure and Activity gait; FMA-LE, Fugl-Meyer Assessment of Lower Extremity; 6MWD, 6-minute Walking Distance; FIM, Functional Independence Measurement; UPDRS-Ⅲ, Unified Parkinson's Disease Rating Scale, Part Ⅲ; BBS, Berg Balance Scale; TUGT, Timed Up and Go Test; 10MWT, 10-meter Walk Test; POMA, Performance Oriented Mobility Assessment; Tinetti POMA, Tinetti Performance Oriented Mobility Assessment; FOG, Freezing of Gait.

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Methodological quality and risk of bias

The methodological quality and risk of bias for each included study are presented in Figures 2 and 3. Four studies [18,19,21,22] reported the random sequence-generation method, while one study [27] used the blinded outcome assessors. All these studies were judged to have low risk of bias in the aforementioned domains. In addition, one study [24] was grouped according to the way patients voluntarily participated. Hence, this was rated as "high risk". All selective reporting and other bias risks included in the trial were considered as "low risk".

Meta-analysis

Effect of Baduanjin on motor function

Unified Parkinson's Disease Rating Scale Part III

UPDRS-III is part of the standard clinical test to assess the severity of motor complications in PD, with higher scores indicating more severe motor complications.[29] Related studies have shown that a change of >3.25 points before and after the intervention may be clinically important for PD.[30] Five studies [19,20,23,24,27] with 418 participants used UPDRS-III to evaluate the motor symptoms. The meta-analysis revealed that the UPDRS-III scores were lower in the Baduanjin group, when compared to the control group (MD: -5.37, 95% CI: -8.96 to -1.78, P=0.003, I^2=94%; Fig. 4a). The heterogeneity sensitivity analysis was performed after excluding the study conducted by Shi *et al.* [19] The subsequent meta-analysis yielded the following results: MD: -3.40, 95% CI: -4.15 to -2.64, P<0.00001, I^2=24% (Fig. 4b). The heterogeneity may be due to the different degrees of PD among participants, and merely the study conducted by Shi *et al.* did not report the stages of the

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participant's PD. These above results indicate that Baduanjin can improve the degree of motor function impairment in PD.

Furthermore, subgroup analyses were performed, according to the intervention durations. The meta-analysis of two studies [20,24] with a Baduanjin intervention duration of ≤8 weeks revealed the following: MD: -2.83, 95% CI: -3.82 to -1.83, P<0.00001, I^2=0%. The meta-analysis of two studies [23,27] with a Baduanjin intervention duration of >8 weeks revealed the following: MD: -4.17, 95% CI: -5.32 to -3.01, P<0.00001, I^2=0% (Fig. 4c). These results indicate that Baduanjin can reduce the degree of motor function impairment in PD, regardless of whether the intervention duration is ≤8 weeks or >8 weeks.

Fugl-Meyer Assessment of Lower Extremity

FMA-LE is a frequently used injury-based measure to assess the motor recovery in the lower extremity and predict the functional recovery of individuals with PD, with higher scores indicative of lower levels of impairment.[31] Scoring is done on a 3-point ordinal scale ranges from 0 (no performance) to 2 (faultless performance). The total score range from 0 (no motor function) to 34 (good motor recovery).[32] Three studies [19,20,22] that involved 313 participants used FMA-LE to assess the lower extremity motor function. The meta-analysis revealed that the FMA-LE scores were higher in the Baduanjin group, when compared to the control group (MD: 5.39, 95% CI: 2.71 to 8.07, P<0.0001, I^2=92%; Fig. 5a). Furthermore, a heterogeneity sensitivity analysis was performed after excluding the study conducted by Song *et al.* [20], and the subsequent meta-analysis results were consistent, with the following results: MD: 6.64, 95% CI: 5.69 to 7.59, P<0.00001, I^2=39% (Fig. 5b). The heterogeneity among these studies may be attributable to the difference in Baduanjin intervention duration.

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These results indicate that Baduanjin can improve the motor function of the lower limbs in PD patients.

Effect of Baduanjin on balance function

Berg Balance Scale

BBS is a test used to assess functional balance, showing high validity and reliability in PD, with higher scores indicating better balance, with scores interpreted in the following manner: 0 to 20, wheelchair bound; 21 to 40, walking with assistance; and 41 to 56, independent.[33,34] Three studies [19,25,27] that included 318 participants used BBS to evaluate the balance function. The meta-analysis revealed that the BBS scores were higher in the Baduanjin group, when compared to the control group (MD: 4.40, 95% CI: 3.08 to 5.73, P<0.00001, I^2=87%; Fig. 6a). A heterogeneity sensitivity analysis was performed after excluding the study conducted by Shi *et al.* [19], and the consistent results were, as follows: MD: 3.71, 95% CI: 2.90 to 4.51, P<0.00001, I^2=63% (Fig. 6b). These results indicate that Baduanjin has a positive effect on balance function in PD.

Effect of Baduanjin on gait

Six-minute walk distance

The 6MWD is a validated and widely used test to assess gait by instructing patients to walk for six minutes, and recording the longest distance the subject could walk, generally suggesting that a change of 14.0 to 30.5 m before and after the intervention may be clinically important for PD.[35,36] Five studies [19-22,27] that involved 485 participants used the 6MWD to evaluate the walking ability. The results revealed that the 6MWD scores were higher in the Baduanjin group, when compared

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to the control group (MD: 21.62, 95% CI: 11.14 to 32.10, P<0.0001, I^2=80%; Fig. 7a). After removing the studies conducted by Zhou *et al.* [22] and Xiao *et al.* [27], the heterogeneity sensitivity analysis and subsequent meta-analysis revealed similar results: MD: 18.86, 95% CI: 13.69 to 24.02, P<0.00001, I^2=0% (Fig. 7b). The heterogeneity may have been attributed to the differences in duration of the Baduanjin interventions between the two studies [22,27] and other studies. These results indicate that Baduanjin can improve the gait of PD patients.

Descriptive analysis

One study [26] presented the data in median (range). Hence, no combined analysis was conducted. The results of this study revealed that Baduanjin can improve the balance and lower limb strength of PD patients. Furthermore, two studies [23,26] used TUGT to evaluate the gait. However, due to the high heterogeneity and small number of pooled studies, the source of heterogeneity could not be identified. Hence, a meta-analysis was not conducted. One study [18] used IDEEA gait, and other study [25] used Tinetti POMA. In addition, one study [26] used FOG, and another one study [23] used the 10MWT. All these methods were used to assess the gait function, and the results suggest that Baduanjin can improve the gait function of PD patients. In addition, the study conducted by Zhang *et al.* [21] used FIM to assess the functional independence, and the results indicated that Baduanjin has a positive effect in improving the functional exercise capacity of PD patients. The study conducted by Dong *et al.* [18] used the Balance-check® balancer test to assess the balance, and the results revealed that Baduanjin can significantly improve the balance of PD patients.

Among the 10 included studies, no adverse events were reported, indicating that Baduanjin is a relatively safe exercise for PD patients. In order to ensure that patients

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adhered to the intervention program, except for one study [26] that did not mention the specific measures, the remaining studies conducted centralized training through coaches during hospitalization or the supervision of family members at home after discharge, and the completion status was reported daily to ensure patient completion of the original protocol.

Publication bias

According to the indicators of UPDRS-III, FMA-LE, BBS and 6MWD, the funnel chart was drawn, and the publication bias test was carried out. The results revealed that the figure for each indicator was not completely symmetrical between the left and right, suggesting that the studies may have a certain degree of publication bias (Fig. 8).

Discussion

The present study revealed that Baduanjin can reduce the UPDRS-III score, and increase the FMA-LE, BBS and 6MWD scores, indicating that Baduanjin might have a positive effect in improving the motor function, balance function, and gait of PD patients. In addition, the subgroup analysis for UPDRS-III revealed an improvement in motor function for PD patients, regardless of whether the duration of intervention was ≤8 weeks or >8 weeks.

Gaps in previous reviews

The present findings are contrary to Hao's review, which explored the effects of 10 different exercise interventions on motor function in PD, and revealed that Baduanjin has no benefit for PD patients. This may be correlated to the limited

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number of studies included in that study (merely three studies).[37] In addition, the conclusions of the present review were similar to those reported by another review, in which Baduanjin practice is beneficial for balance and trunk flexibility. However, that study included a population of healthy or people with different types of diseases.[38]

Effects of Baduanjin on motor function, balance function and gait

As a chronic disease, PD has a long course, and is prone to recurrence, which is mainly caused by the progressive degeneration of dopaminergic neurons in the substantia nigra pars compacta, the decrease in striatal dopamine, and the presence of Lewy bodies.[39] Multiple abnormal neural circuits can lead to balance and gait problems in PD patients.[40] Gait problems are strongly dependent on impaired balance control, and the poor control of both can cause movement disorders.[41,42] Levodopa is the main drug for relieving motor symptoms of PD. However, approximately 50% of patients who receive levodopa for long durations would develop levodopa-induced dyskinesia (LID).[43] Speck *et al.* [44] suggested through a mice model that exercise can partially prevent the development of LID, and attenuate the side effects of levodopa through the normalization of striatopallidal dopaminergic signaling, without affecting the anti-parkinsonian effect of levodopa. Hence, this indicates that Baduanjin may induce an effect that enhances the medication efficiency and ameliorates the side effects of levodopa, making this a good CAM for the treatment of PD.

Several large controlled clinical studies have revealed that continuous exercise can improve the performance of daily activities in the early stage of PD, such as balance, gait and movement function, which is consistent with the present findings.[45,46] The exercise guidelines for PD recommends 20-60 minutes of

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moderate-intensity aerobic exercise for 3-5 days a week.[47] Baduanjin, as a moderate intensity aerobic exercise, mainly comprises of eight simple left-right symmetrical movements, such as arm raising, shaking of the head, waist rotation and foot flipping.[48] Practicing Baduanjin can fully stretch the joints of the upper and lower limbs, and the movement of the trunk can improve the ability of PD patients to control their center of gravity. At the same time, Baduanjin can also improve the proprioception, strengthen the coordination and sensitivity of limbs, and promote the recovery of balance function.[49] Ye *et al.* [50] reported that the upper limb exercise of Baduanjin can enhance the patient's coordination, and the knee flexion and squat movement can enhance the lower limb flexor and extensor muscle strength, and lower limb weight-bearing capacity, further improving the gait ability. This explains the findings reported in the present review.

The muscle tension caused by the uncoordinated contraction of active and antagonistic muscles influence a series of motor symptoms in patients with PD.[51] There is evidence that mind-body exercises, including Tai Chi, Yoga and Qigong, can reduce the contraction of antagonist muscles.[52,53] As a kind of mind-body exercise, the long-term practice of Baduanjin can promote blood circulation in the upper and lower limbs, and lower back, strengthened the muscle metabolism and stress function, improve the synergistic contraction of active muscles and antagonist muscles, and promote the recovery of motor function.[50] The α-syn protein is the main pathogenic protein of PD, and its accumulation is the main reason of neuronal loss.[54] Existing studies have revealed that aerobic exercise can reduce the loss of dopaminergic neurons, increase synaptic connections, and upregulate neurotrophic factor levels to improve PD dyskinesia. Furthermore, this can downregulate the α-Syn protein levels and neuronal apoptosis, which in turn, can reduce inflammation and mitochondrial

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dysfunction to restore motor function in PD patients.[55-57] Al-Jarrah *et al.* [58] proposed that aerobic exercise can decrease the level of proinflammatory proteins in the striatum in a PD experimental model by reducing the activity of microglia, thereby alleviating PD symptoms. These might be the possible mechanisms for the effect of Baduanjin in improving motor function.

Limitations and future prospects

The present systematic review has several limitations. First, the present study included PD patients with Hoehn and Yahr grades of below grade 3 or no reported grade. Hence, the results cannot be extended to PD patients with Hoehn and Yahr grades 4-5. Second, the present study had a relatively high heterogeneity among the literatures, such as different study intervention durations, which might have been the cause for the clinical heterogeneity. Third, merely databases in the English and Chinese language were searched. Hence, it is possible that some researches may have been missed during the database search.

Conclusions

The present review revealed the potential benefits of Baduanjin in improving the motor function, balance function, and gait ability of PD patients. Therefore, Baduanjin exercise may be taken as a valuable aerobic exercise in clinical rehabilitation for the application of CAM. However, all findings must be interpreted with caution due to limitations in both the quantity and quality of available evidence. Furthermore, additional large, rigorous trials are warranted to better characterize the effects of Baduanjin in PD.

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Authors' contributions

Jinghui Lai, Yangfan Cai and Liyan Yang conceived and designed the study. Jinghui Lai drafted the manuscript. Yangfan Cai and Min Xia performed the data collection. Jinghui Lai, Xi Cheng and Ying Chen performed the data analysis. All authors approved the final manuscript before submission.

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Conflicts of interest

The authors declare that they have no conflict of interest.

Data sharing statement

All data relevant to the study are included in the article or uploaded as supplementary information.

Ethics approval

The study did not involve human participants.

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Figure legends

Figure 1 Literature screening process.

Figure 2 Risk of bias graph.

Figure 3 Risk of bias summary.

Figure 4 (a) Total UPDRS-III scores after the Baduanjin exercise. (b) Total UPDRS-III scores after the Baduanjin exercise (sensitivity analysis). (c) Subgroup analysis of the UPDRS-III scores after the Baduanjin exercise, according to intervention duration.

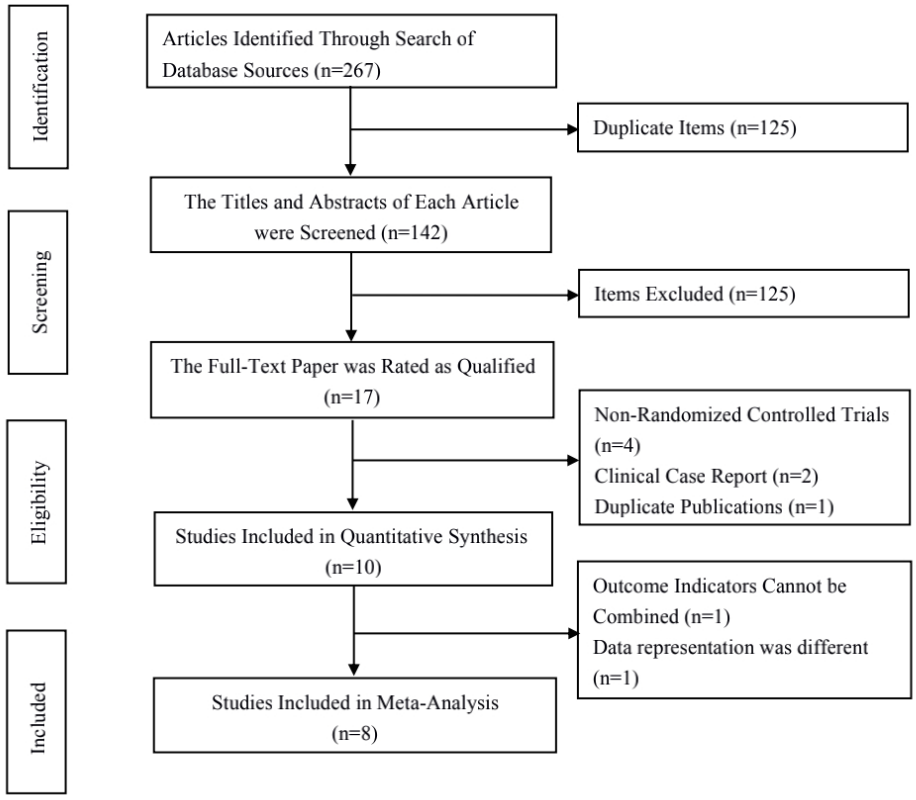
Figure 5 (a) Total FMA-LE scores after the Baduanjin exercise. (b) Total FMA-LE scores after the Baduanjin exercise (sensitivity analysis).

Figure 6 (a) Total BBS scores after the Baduanjin exercise. (b) Total BBS scores after the Baduanjin exercise (sensitivity analysis).

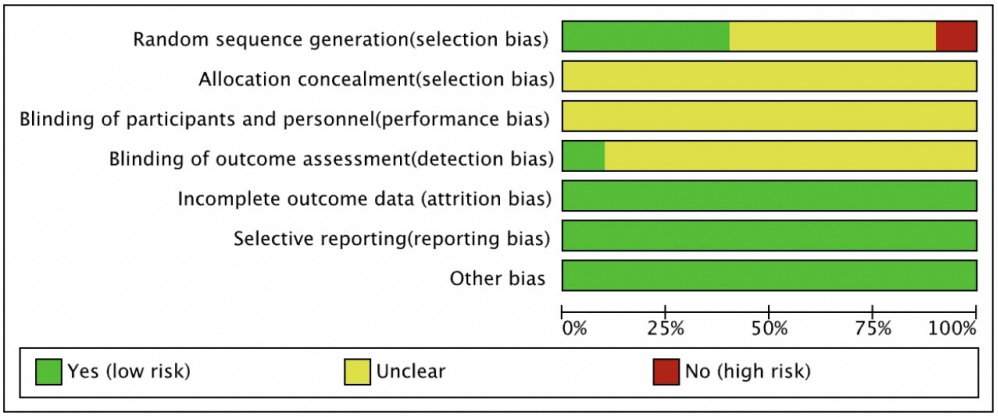
Figure 7 (a) Total 6WMD scores after the Baduanjin exercise. (b) Total 6WMD scores after the Baduanjin exercise (sensitivity analysis).

Figure 8 (a) Funnel chart for UPDRS-III. (b) Funnel chart for FMA-LE. (c) Funnel chart for BBS. (d) Funnel chart for 6MWD.

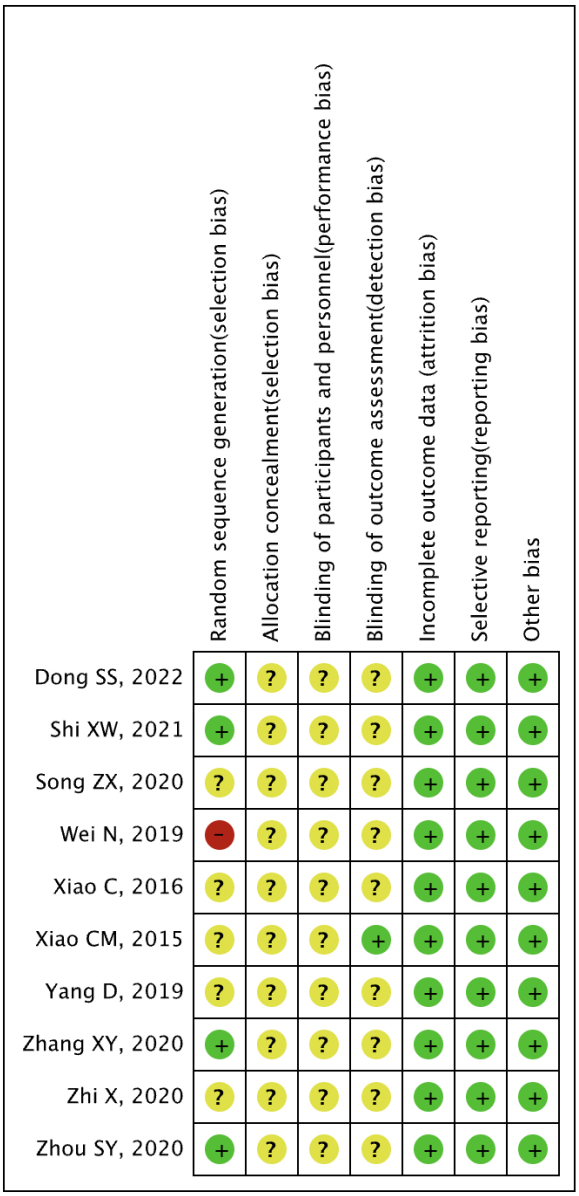
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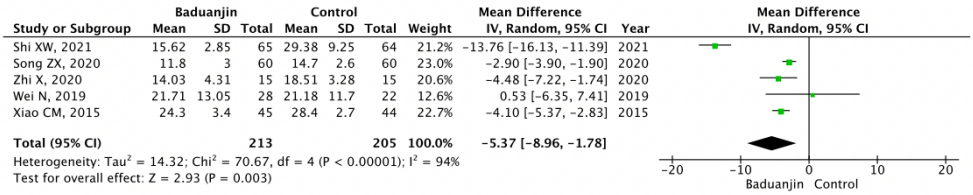
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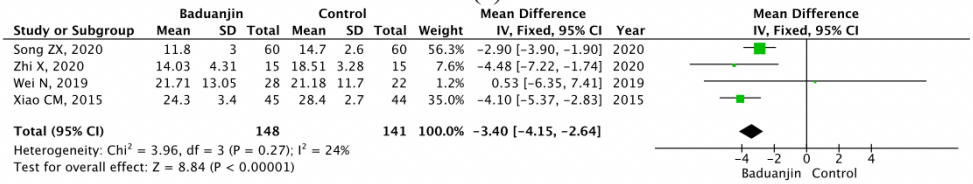
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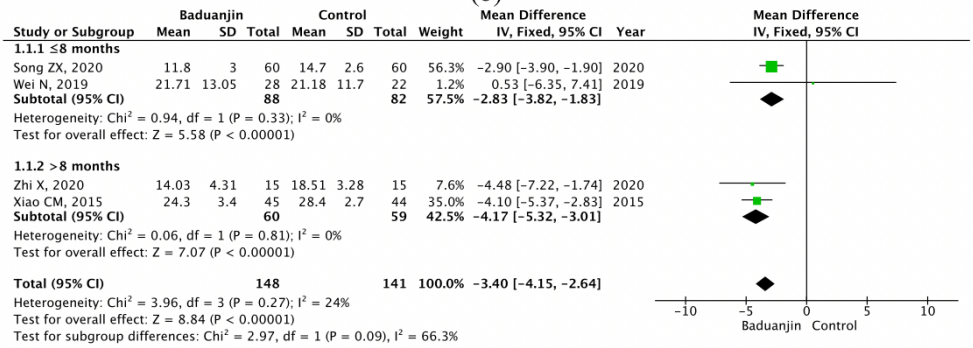
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(a)

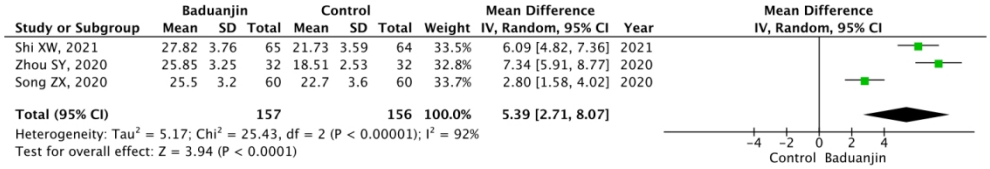


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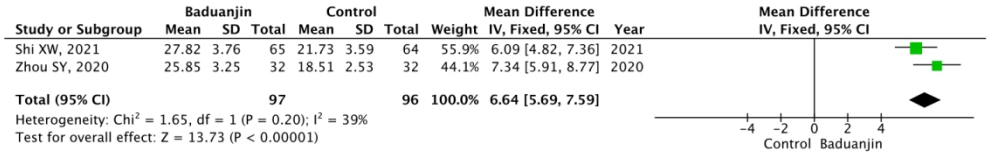


(c)

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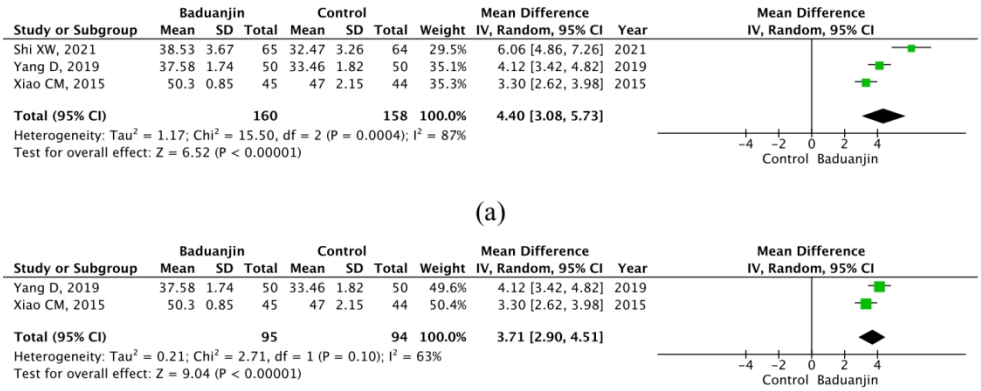


(a)



(b)

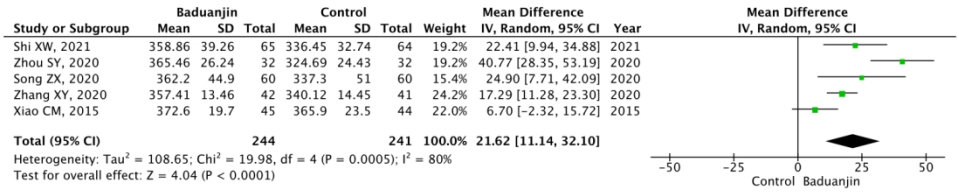
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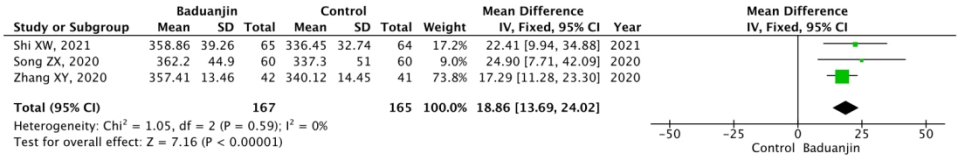
(a)

(b)

280x131mm (144 x 144 DPI)

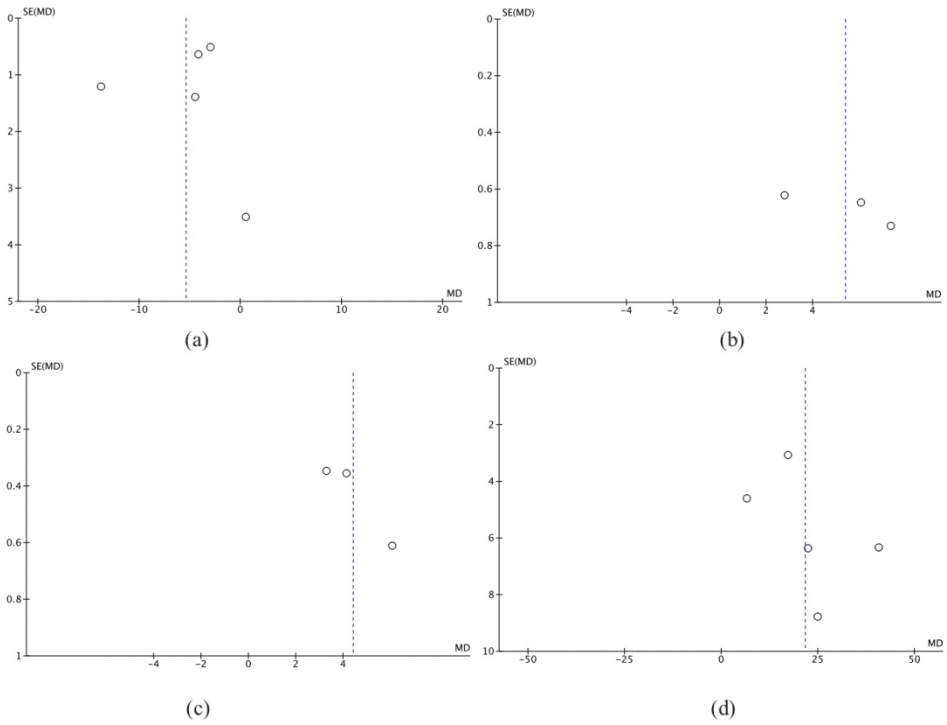


(a)



(b)

287x145mm (144 x 144 DPI)



277x201mm (144 x 144 DPI)

Search strategy

PubMed

#1 Qigong[MeSH Terms] Filters: Chinese, English.

#2 (((Baduanjin[Title/Abstract]) OR (Mind-body exercise[Title/Abstract])) OR (Ch'i Kung[Title/Abstract])) AND (traditional exercises[Title/Abstract]) Filters: Chinese, English.

#3 #1 OR #2

#4 (Parkinson disease[MeSH Terms]) OR (Parkinsonian Disorders[MeSH Terms]) Filters: Chinese, English.

#5 ((Parkinsonian Diseases[Title/Abstract]) OR (Parkinsonian Syndrome[Title/Abstract])) OR (PD[Title/Abstract]) Filters: Chinese, English.

#6 #4 OR #5

#7 #3 AND #6

((Qigong[MeSH Terms]) OR ((((Baduanjin[Title/Abstract]) OR (Mind-body exercise[Title/Abstract])) OR (Ch'i Kung[Title/Abstract])) OR (traditional exercises[Title/Abstract]))) AND (((Parkinson disease[MeSH Terms]) OR (Parkinsonian Disorders[MeSH Terms])) OR (((Parkinsonian Diseases[Title/Abstract]) OR (Parkinsonian Syndrome[Title/Abstract])) OR (PD[Title/Abstract]))) Filters: Chinese, English.

Web of Science

#1 ((((TS=(Qigong)) OR TS=(Baduanjin)) OR TS=(Mind-body exercise)) OR TS=(Ch'i Kung)) OR TS=(traditional exercises)

#2 ((((TS=(Parkinson disease)) OR TS=(Parkinsonian Disorders)) OR TS=(Parkinsonian Diseases)) OR TS=(Parkinsonian Syndrome)) OR TS=(PD)

#3 #1 AND #2

((((TS=(Qigong)) OR TS=(Baduanjin)) OR TS=(Mind-body exercise)) OR TS=(Ch'i Kung)) OR TS=(traditional exercises) AND ((((TS=(Parkinson disease)) OR TS=(Parkinsonian Disorders)) OR TS=(Parkinsonian Diseases)) OR TS=(Parkinsonian Syndrome)) OR TS=(PD) AND English OR Chinese (Languages)

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Cochrane Library

#1 MeSH descriptor: [Qigong] explode all trees

#2 (Baduanjin OR Mind-body exercise OR Ch'i Kung OR traditional exercises):ti,ab,kw

#3 #1 OR #2

#4 MeSH descriptor: [Parkinson Disease] explode all trees

#5 MeSH descriptor: [Parkinsonian Disorders] explode all trees

#6 (Parkinsonian Diseases OR Parkinsonian Syndrome OR PD):ti,ab,kw

#7 #4 OR #5 OR #6

#8 #3 AND #7

Embase

#1 'qigong'/exp AND ([chinese]/lim OR [english]/lim)

#2 ('baduanjin':ab,ti OR 'mind-body exercise':ab,ti OR 'chi kung':ab,ti OR 'traditional exercises':ab,ti) AND ([chinese]/lim OR [english]/lim)

#3 #1 OR #2

#4 'parkinson disease'/exp OR 'parkinsonian disorders'/exp AND ([chinese]/lim OR [english]/lim)

#5 ('parkinsonian diseases':ab,ti OR 'parkinsonian syndrome':ab,ti OR pd:ab,ti) AND ([chinese]/lim OR [english]/lim)

#6 #4 OR #5

#7 #3 AND #6

EBSCOhost

#1 SU qigong Language:- english or chinese

#2 TX (Baduanjin OR Mind-body exercise OR Ch'i Kung OR traditional exercises) Language:- english or chinese

#3 #1 OR #2 SU qigong OR TX (Baduanjin OR Mind-body exercise OR Ch'i Kung OR traditional exercises)

#4 SU (parkinson disease OR parkinsonian disorders) Language:- english or chinese

#5 TX (Parkinsonian Diseases OR Parkinsonian Syndrome OR PD) Language:- english or chinese

#6 #4 OR #5 SU (parkinson disease OR parkinsonian disorders) OR (Parkinsonian Diseases OR

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Parkinsonian Syndrome OR PD)

#7 #3 AND #6

OVID

#1 qigong.hw. limit (chinese or english)

#2 (Baduanjin or Mind-body exercise or Ch'i Kung or traditional exercises).af. limit (chinese or english)

#3 #1 or #2 qigong.hw. or (Baduanjin or Mind-body exercise or Ch'i Kung or traditional exercises).af.

#4 (parkinson disease or parkinsonian disorders).hw. limit (chinese or english)

#5 (Parkinsonian Diseases or Parkinsonian Syndrome or PD).af. limit (chinese or english)

#6 #4 or #5 (parkinson disease or parkinsonian disorders) or (Parkinsonian Diseases or Parkinsonian Syndrome or PD).af.

#7 #3 and #6

SinoMed

#1 ("气功"[加权:扩展])

#2 "八段锦"[中文标题:智能] OR "传统功法"[中文标题:智能] OR "身心运动"[中文标题:智能] OR "体育疗法"[中文标题:智能]

#3 "八段锦"[摘要:智能] OR "传统功法"[摘要:智能] OR "身心运动"[摘要:智能] OR "体育疗法"[摘要:智能]

#4 #1 OR #2 OR #3

#5 "帕金森病"[加权:扩展]

#6 "帕金森综合征"[中文标题:智能] OR "帕金森障碍"[中文标题:智能]

#7 "帕金森综合征"[摘要:智能] OR "帕金森障碍"[摘要:智能]

#8 #5 OR #6 OR #7

#9 #4 AND #8

China National Knowledge Infrastructure (CNKI)

#1 SU=八段锦+传统功法+身心运动+气功+体育疗法

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#2 SU=帕金森病+帕金森综合征+帕金森障碍

#3 #1 and #2

(SU=八段锦+传统功法+身心运动+气功+体育疗法) and (SU=帕金森病+帕金森综合征+帕金森障碍)

Wanfang Data

#1 主题:八段锦+传统功法+身心运动+气功+体育疗法

#2 主题:帕金森病+帕金森综合征+帕金森障碍

#3 #1 and #2

(主题:八段锦+传统功法+身心运动+气功+体育疗法) and (主题:帕金森病+帕金森综合征+帕金森障碍)

China Science Journal Database (VIP)

#1 M=八段锦 OR M=传统功法 OR M=身心运动 OR M=气功 OR M=体育疗法

#2 M=帕金森病 OR M=帕金森综合征 OR M=帕金森障碍

#3 #1 AND #2

(M=八段锦 OR M=传统功法 OR M=身心运动 OR M=气功 OR M=体育疗法) AND (M=帕金森病 OR M=帕金森综合征 OR M=帕金森障碍)

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page2
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page5
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page6
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page6
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	No
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page6
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page6
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	No
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page5

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	No
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Page7
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page9
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Figure 4-7
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page9-11
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Page9-11
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page9-11
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	No
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page9
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page9-11
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page11-13
	23b	Discuss any limitations of the evidence included in the review.	Page13
	23c	Discuss any limitations of the review processes used.	Page13
	23d	Discuss implications of the results for practice, policy, and future research.	Page13
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	No
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	No
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page14
Competing interests	26	Declare any competing interests of review authors.	Page14
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	No

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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